Numerical-relativity particle-in-cell method at low particle number: An independent reconstruction by language-model agents

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Provisional abstract of a draft in progress; results marked [pending] were not final when it was written. Collisionless matter near black holes obeys the Einstein-Vlasov system, where particles follow geodesics of their curvature. Wu and Most introduced a numerical-relativity particle-in-cell method with millions of phase-space packets called macroparticles. We ask whether their text enables an independent rebuild and what changes at ten thousand macroparticles. Language-model agents under human direction built the code from text alone in AthenaK, with separate models for code, checks, and writing. Predictions written before each control run tested every diagnosis. Each step evolves spacetime like a field solver, advances macroparticles by a pusher, and spreads sources onto cells by cloud-in-cell deposition. The implicit midpoint pusher matches Kerr orbits, while the spacetime Boris pusher needs an unstated local-frame tetrad gauge: symmetric analytically, Cholesky on meshes. Tracers near fixed and evolving black holes conserve invariants, and deposition conserves sources across refinement. Geometric slice-mass checks of apparent horizons, outermost surfaces of non-expanding light rays, exposed an unfilled ghost-cell defect in AthenaK's apparent-horizon finder. Interpolant derivatives agree within the accuracy of the mass measure. Surface mass rescaling removes dust-collapse lattice excess at ten thousand macroparticles. A filter compensation pass cancels cloud widening that expands stable clusters. Level-capped deposition coarsens clouds yet worsens constraints and apparent-horizon mass drift near black holes, so finest-level deposition stays. Dust collapses to a black hole of matching apparent-horizon mass while lapse excision removes interior particles frozen by time dilation. Monoenergetic clusters, equal-energy stellar spheres, stay bound with a growing core oscillation. The cause of that growth and coarsened-cloud drift is not determined. Past the binding-energy maximum, unperturbed unstable clusters expand toward migration [pending], perturbed ones collapse [pending], and unbound ones disperse [pending]. Prolate clouds collapse [pending], and a relativistic cluster with two collapsing overdensities at speed-consistent radius merges [pending]. These controls establish a reproducible algorithm: ten thousand macroparticles capture black hole formation and cluster stability.

I. INTRODUCTION

Stars in dense relativistic clusters and collisionless dark matter around black holes move freely through collective gravitational fields that vary strongly in space and time [1–5]. Two-body gravitational encounters act over timescales far longer than a dynamical crossing time, much as binary collisions remain negligible in a collisionless plasma [2, 6]. Multi-stream trajectories cross in configuration space and build anisotropic stresses, whereas single-fluid hydrodynamics assigns one bulk velocity to each spatial point [3, 7]. Strong-field evolution therefore requires the Einstein-Vlasov system, a phase-space kinetic description coupled to the full Einstein equations where matter follows geodesics of the curvature its stress-energy generates [3, 8, 9]. Early numerical studies with spherical codes mapped equilibrium sequences, radial stability, and black hole formation in relativistic clusters [1, 10–12] alongside critical gravitational collapse [13–15]. Subsequent axisymmetric and three-dimensional simulations extended kinetic gravity to rotating clusters, nonspherical collapse, and cosmic structure [7, 16–20], where asymmetric instabilities and mergers demand evolution in three spatial dimensions.

The numerical-relativity particle-in-cell (NRPIC) method of Ref. [21] brings algorithms from relativistic plasma simulation [6, 22–25] to three-dimensional Einstein-Vlasov evolution. Macroparticles—finite-mass packets sampling six-dimensional phase space—advance along timelike geodesics through a time-reversible trajectory pusher in the reconstructed metric [21, 26–29]. Conservative cloud-in-cell deposition assigns each macroparticle's energy, momentum, and stress to nearby mesh cells across refinement boundaries before grid filtering smooths discrete sampling noise [6, 21, 22, 30]. A constraint-damped solver in moving-puncture gauge, the standard way to evolve Einstein's equations through black hole formation on a fixed grid, advances the spacetime on an adaptive mesh [21, 31–34]. Reference [21] demonstrated this coupled cycle on gravitational collapse and relativistic cluster benchmarks using millions of macroparticles on large machines.

Two questions decide how broadly this method can be adopted. One question asks whether an independent builder without access to the original code can reconstruct the NRPIC method quantitatively from its published description alone. A numerical method is usable by others only when its written equations and algorithmic rules suffice for an independent implementation to recover its benchmarks. The second question asks which results survive at ten thousand macroparticles, where a run takes minutes to a few hours on one graphics card, and what must change for that. A small computational entry cost decides who can use the method for routine physical exploration, rapid parameter surveys, and independent verification.

Language-model agents under human direction built an independent implementation from the published text alone inside the public AthenaK code [35, 36], with portable parallel execution through Kokkos [37]. Every algorithmic stage and coupled test was compared with exact analytical solutions or with the published numbers of Ref. [21]. Candidate explanations of unexpected results were tested against predictions recorded before control runs designed so that they could fail, as described in Sec. VI.

The published description suffices to rebuild the method and reproduce the single-particle errors of Ref. [21] once one unstated choice is identified (Sec. IV). That choice is the tetrad gauge, the local rotation freedom of the orthonormal frame in the spacetime Boris pusher, where distinct frame constructions shift the discrete error (Secs. III A and IV). Self-gravitating evolutions at ten thousand macroparticles require two algorithmic changes (Sec. V). Surface masses on a cubic dust lattice must be rescaled because discrete site counts inside a sphere exceed its continuum volume at low particle number (Sec. V A). Multi-pass source filters require a sharpening compensation pass [6] because plain smoothing widens clouds like gravitational softening and makes stable clusters bounce outward (Secs. III C and V C). Coarser particle clouds—a tempting change on sparse refined meshes—raise constraint violations and drive black hole mass drift, so clouds belong on the finest local level (Sec. V). A geometric mass measure of the slice also exposed a ghost-zone defect in the host code's finder for apparent horizons, the outermost closed surfaces of zero outgoing light expansion (Secs. III F and V A). With these elements in place, ten thousand macroparticles reproduce in kind dust collapse [38], the sequence of relativistic clusters [1, 12], and the migration of an unstable cluster toward the stable branch [21, 39] (Secs. V A and V C). Documented differences from Ref. [21] appear in the collapse times of unstable clusters and in the collapse of one perturbed near-critical model (Sec. V C).

Section II formulates collisionless matter in a dynamical spacetime and its macroparticle phase-space representation. Section III presents the particle-mesh method: geodesic pushers, metric reconstruction, stress-energy deposition and filtering, spacetime evolution, adaptive mesh refinement, and numerical methods. Section IV tests the geodesic pushers, phase mixing, source conservation, and interior particle removal by lapse excision around fixed and evolving black holes. Section V studies self-gravitating matter at ten thousand macroparticles across spherical dust collapse and the sequence of relativistic clusters. Section VI describes how the code was assembled and verified, before the final section states our conclusions and the appendix records the scenario parameters.

II. COLLISIONLESS MATTER IN A DYNAMICAL SPACETIME

Stars in dense clusters, particles in dark-matter spikes, and debris in colliding stellar streams move freely through smooth collective gravitational fields [1–3]. Two-body gravitational encounters act over timescales far longer than a dynamical crossing time, so each constituent follows a geodesic of the spacetime geometry generated by the whole ensemble. Throughout this paper we set Newton's gravitational constant G and the speed of light c to unity and adopt metric signature $(-,+,+,+)$. Greek letters μ,ν,ρ,σ label spacetime indices, Latin letters i,j,k,l label spatial coordinate indices, and parenthesized indices $(m),(n),(k)$ label components in a tetrad, a local frame of four mutually perpendicular unit vectors.

Collisionless matter in a curved spacetime with metric $g_{\mu\nu}$ and inverse $g^{\mu\nu}$ is described in phase space by a rest-mass-weighted distribution function $f(x^\mu,u^\nu)$ [3, 8, 9]. Here x^μ are spacetime coordinates and u^μ is the four-velocity, equal to the four-momentum per unit rest mass on the future mass shell $g_{\mu\nu}u^\mu u^\nu=-1$. Phase-space conservation along free-fall worldlines coupled to Einstein's field equations forms the Einstein-Vlasov system,

$$u^\mu\partial_\mu f-\Gamma^\mu_{\nu\rho}u^\nu u^\rho\frac{\partial f}{\partial u^\mu}=0,\qquad G_{\mu\nu}=8\pi T_{\mu\nu},\tag{1}$$

where $\Gamma^\mu_{\nu\rho}$ are the Christoffel symbols of $g_{\mu\nu}$. The Einstein tensor $G_{\mu\nu}$ is sourced by the stress-energy tensor $T_{\mu\nu}$, given by the second four-velocity moment of f over the future mass shell.

Time evolution splits the spacetime into spacelike hypersurfaces of constant coordinate time t with spatial coordinates x^i, induced metric γ_{ij}, inverse γ^{ij}, and determinant $\gamma=\det(\gamma_{ij})$ [40]. Local measurements on each slice are referred to normal observers, Eulerian observers whose worldlines cross the slice orthogonally with future-directed unit four-velocity n^μ and covector $n_\mu=(-\alpha,0,0,0)$. The lapse α links adjacent slices by converting a coordinate interval $\mathrm{d}t$ into proper time $\alpha\,\mathrm{d}t$ along normal observers. The shift β^i gives the coordinate velocity at which spatial grid points drift relative to those normal observers. In these variables the spacetime line element reads

$$\mathrm{d}s^2=-\alpha^2\,\mathrm{d}t^2+\gamma_{ij}(\mathrm{d}x^i+\beta^i\,\mathrm{d}t)(\mathrm{d}x^j+\beta^j\,\mathrm{d}t),\tag{2}$$

with $n^\mu=(\alpha^{-1},-\alpha^{-1}\beta^i)$. The bending of each slice inside the ambient spacetime is measured by the symmetric extrinsic curvature K_{ij}, defined through

$$\partial_t\gamma_{ij}=-2\alpha K_{ij}+D_i\beta_j+D_j\beta_i,\tag{3}$$

where D_i is the spatial covariant derivative compatible with γ_{ij} and $\beta_i=\gamma_{ij}\beta^j$. Normal observers project $T^{\mu\nu}$ into ten matter fields—the energy density E, covariant momentum density S_i, and symmetric spatial stress

S_{ij}—according to

$$E = n_\mu n_\nu T^{\mu\nu}, \qquad S_i = -g_{i\mu} n_\nu T^{\mu\nu}, \\ S_{ij} = g_{i\mu} g_{j\nu} T^{\mu\nu}. \tag{4}$$

Direct grid discretization of f spans six phase-space dimensions and becomes prohibitive in three spatial dimensions without symmetry, whereas the Vlasov equation conserves phase-space weight along timelike geodesics. A particle-in-cell method [6, 21, 22] exploits these characteristics by sampling f with N finite-weight phase-space packets called macroparticles, indexed by $p = 1, \ldots, N$. With $u_\mu = g_{\mu\nu} u^\nu$, each macroparticle carries fixed rest mass m_p, spatial coordinates $x^i_p(t)$, and covariant spatial four-velocity components $u_{i,p}(t) = g_{i\mu} u^\mu_p$, giving six phase-space variables per particle. On the future mass shell $g^{\mu\nu} u_\mu u_\nu = -1$, the Lorentz factor seen by normal observers is

$$W = \alpha u^0 = \sqrt{1 + \gamma^{ij} u_i u_j}, \tag{5}$$

which determines the single-particle Hamiltonian $-u_0 = \alpha W - \beta^i u_i$. Hamilton's canonical equations for (x^i, u_i) yield the characteristic equations,

$$\frac{\mathrm{d}x^i}{\mathrm{d}t} = \frac{\alpha \gamma^{ij} u_j}{W} - \beta^i, \\ \frac{\mathrm{d}u_i}{\mathrm{d}t} = -W \partial_i \alpha + u_j \partial_i \beta^j - \frac{\alpha}{2W} u_j u_k \partial_i \gamma^{jk}. \tag{6}$$

At spatial coordinate $\boldsymbol{x} = (x^1, x^2, x^3)$, the ensemble stress-energy projected onto normal observers forms the particle sources,

$$E(t, \boldsymbol{x}) = \sum_{p=1}^{N} \frac{m_p W_p}{\sqrt{\gamma}} \, \delta^3(\boldsymbol{x} - \boldsymbol{x}_p), \\ S_i(t, \boldsymbol{x}) = \sum_{p=1}^{N} \frac{m_p u_{i,p}}{\sqrt{\gamma}} \, \delta^3(\boldsymbol{x} - \boldsymbol{x}_p), \\ S_{ij}(t, \boldsymbol{x}) = \sum_{p=1}^{N} \frac{m_p u_{i,p} u_{j,p}}{W_p \sqrt{\gamma}} \, \delta^3(\boldsymbol{x} - \boldsymbol{x}_p). \tag{7}$$

Here $\delta^3(\boldsymbol{x} - \boldsymbol{x}_p)$ is the three-dimensional coordinate Dirac distribution and W_p is evaluated at $\boldsymbol{x}_p(t)$. Geodesic transport of $(x^i_p, u_{i,p})$ under the characteristic equations, coupled to smoothed mesh projections of the particle sources, solves the Einstein-Vlasov system.

III. THE PARTICLE-MESH METHOD

Coupled Einstein-Vlasov evolution advances smooth curvature fields on an Eulerian mesh alongside discrete trajectories over a shared global time step Δt, set by the Courant limit of the finest cell spacing Δx. The numerical-relativity particle-in-cell (NRPIC) method [21] links both representations in a staggered cycle, analogous to electromagnetic particle-in-cell codes where Maxwell's equations take the place of the spacetime solver [6, 22]. Finite sampling with a mean count N_{ppc} of macroparticles per cell introduces sampling noise, discrete fluctuations of relative amplitude $N_{\mathrm{ppc}}^{-1/2}$. Each stage of the cycle preserves orbital time-reversal symmetry, conserves deposited weight across refinement boundaries, and keeps grid sources smooth at modest N_{ppc}.

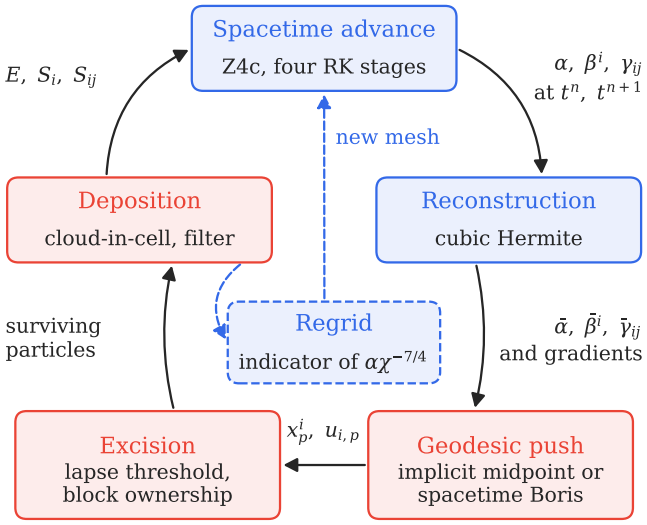


FIG. 1. One step of the particle-mesh cycle from t^n to t^{n+1}. Blue stages act on the mesh, red stages on the macroparticles. Arrows state the quantities passed; overbars mark metric fields averaged over the step. On the dashed regrid branch, metric fields, block ownership, and sources are rebuilt on the new mesh.

One cycle from t^n to $t^{n+1} = t^n + \Delta t$ runs through six stages in sequence (Fig. 1). First, α, β^i, and γ_{ij} at t^n are saved across all active and ghost cells. Second, four fourth-order Runge–Kutta stages advance the spacetime variables to t^{n+1} while holding the sources (E, S_i, S_{ij}) from t^n fixed. Third, a geodesic pusher—an update advancing each macroparticle's position and momentum across Δt—pushes every particle from t^n to t^{n+1} in the metric averaged between both endpoints. Fourth, lapse excision removes particles deep inside black holes where $\alpha(t^n, \boldsymbol{x}^n_p) < \alpha_{\mathrm{exc}}$ for a threshold α_{exc}. Out-of-domain or nonfinite states are discarded, and surviving particles are reassigned to their enclosing blocks. Fifth, stress-energy deposition, the assignment of each macroparticle's mass-energy, momentum, and stress to nearby cells, deposits (E, S_i, S_{ij}) at t^{n+1}, filters the ten arrays, and evaluates the constraints. Sixth, diagnostics run and the adaptive mesh refines or coarsens where curvature gradients cross thresholds. After any regrid, metric arrays, block ownership, and sources are rebuilt on the new leaf cells while $(m_p, x^i_p, u_{i,p})$ stay unchanged.

A. Geodesic pushers

Collisionless equilibria and close black hole passages require trajectory updates free of secular drift in orbital invariants. The implicit midpoint pusher and the spacetime Boris pusher [21] store x_p^i and $u_{i,p}$ at integer time levels, reach second-order accuracy in Δt, and preserve time-reversal symmetry through midpoint evaluation. Each queries the geometry solely through a point-evaluation function returning α, β^i, γ_{ij}, and their first spatial derivatives, so identical code runs on analytic metrics, uniform lattices, and adaptive meshes. Fixed-point iteration updates the six endpoint variables $(x_p^{i,n+1}, u_{i,p}^{n+1})$ until the largest component change falls below a tolerance.

The implicit midpoint pusher evaluates the characteristic equations at the phase-space midpoint (with the particle index p suppressed),

$$\bar{x}^i = \frac{x^{i,n} + x^{i,n+1}}{2}, \qquad \bar{u}_i = \frac{u_i^n + u_i^{n+1}}{2}. \tag{8}$$

Spatial reconstructions at t^n and t^{n+1} are evaluated at $\bar{\boldsymbol{x}}$ and averaged to obtain $\bar{\alpha}$, $\bar{\beta}^i$, $\bar{\gamma}_{ij}$, and their spatial derivatives. Matrix inversion of $\bar{\gamma}_{ij}$ yields $\bar{\gamma}^{ij}$, $\bar{u}^i = \bar{\gamma}^{ij}\bar{u}_j$, and $\bar{W} = \sqrt{1 + \bar{\gamma}^{ij}\bar{u}_i\bar{u}_j}$, advancing the trajectory by the midpoint update,

$$\begin{aligned} \frac{x^{i,n+1} - x^{i,n}}{\Delta t} &= \frac{\bar{\alpha}\bar{\gamma}^{ij}\bar{u}_j}{\bar{W}} - \bar{\beta}^i, \\ \frac{u_i^{n+1} - u_i^n}{\Delta t} &= -\bar{W}\partial_i\bar{\alpha} + \bar{u}_j\partial_i\bar{\beta}^j + \frac{\bar{\alpha}}{2\bar{W}}\bar{u}^j\bar{u}^k\partial_i\bar{\gamma}_{jk}, \end{aligned} \tag{9}$$

where $\bar{u}^j\bar{u}^k\partial_i\bar{\gamma}_{jk} = -\bar{u}_j\bar{u}_k\partial_i\bar{\gamma}^{jk}$. Fixed by the coordinate metric alone, the midpoint update conserves any momentum u_i conjugate to a cyclic coordinate ($\partial_i = 0$) to rounding error [27].

The spacetime Boris pusher projects momentum onto the normal-observer tetrad, pairing $e^\mu_{(0)} = n^\mu$ with a spatial triad $e_i^{(m)}$ of three mutually perpendicular unit covectors on the slice [21, 28, 29]. The triad and its inverse $e^i_{(m)}$ satisfy

$$\gamma_{ij} = \delta_{mn}e_i^{(m)}e_j^{(n)}, \qquad e^i_{(m)}e_i^{(n)} = \delta^n_m, \tag{10}$$

with Kronecker deltas δ_{mn} and δ^n_m. Projection defines the tetrad momentum $u_{(m)} = e^i_{(m)}u_i$ (so $u_i = e_i^{(n)}u_{(n)}$) and $W = \sqrt{1 + \delta^{mn}u_{(m)}u_{(n)}}$. Along a trajectory with $u^j = \gamma^{jk}u_k$, coordinate velocity v^j, and convective derivative

$$\frac{\mathrm{d}}{\mathrm{d}t} = \partial_t + v^j\partial_j, \qquad v^j = \frac{\alpha u^j}{W} - \beta^j, \tag{11}$$

differentiating $u_i = e_i^{(n)}u_{(n)}$ gives

$$e_i^{(n)}\frac{\mathrm{d}u_{(n)}}{\mathrm{d}t} = \frac{\mathrm{d}u_i}{\mathrm{d}t} - u_{(n)}\big(\partial_t e_i^{(n)} + v^j\partial_j e_i^{(n)}\big). \tag{12}$$

Spatial differentiation of $\gamma_{jk} = \delta_{mn}e_j^{(m)}e_k^{(n)}$ rewrites the metric-gradient term in the characteristic equations as

$$\frac{1}{2}u^j u^k\partial_i\gamma_{jk} = u^j u_{(n)}\partial_i e_j^{(n)}. \tag{13}$$

With the momentum-contracted triad gradients

$$c_{jk} = u_{(n)}\partial_j e_k^{(n)}, \qquad c_k^t = u_{(n)}\partial_t e_k^{(n)}, \tag{14}$$

inserting the characteristic equations, contracting with $e^i_{(m)}$, and passing to normal-observer proper time $\mathrm{d}\hat{t} = \alpha\,\mathrm{d}t$ yield the tetrad equation of motion,

$$\frac{\mathrm{d}u_{(m)}}{\mathrm{d}\hat{t}} = \tilde{E}_{(m)} + \frac{1}{W}\epsilon_{(m)(n)(k)}u_{(n)}\tilde{B}_{(k)}, \tag{15}$$

where $\epsilon_{(m)(n)(k)}$ is the Levi-Civita symbol. Here the momentum-contracted gravito-electric and gravitomagnetic fields $\tilde{E}_{(m)}$ and $\tilde{B}_{(k)}$ read

$$\begin{aligned} \alpha\tilde{E}_{(m)} &= e^i_{(m)}\big(-W\partial_i\alpha + u_j\partial_i\beta^j + \beta^j c_{ji} - c_i^t\big), \\ \tilde{B}_{(k)} &= \epsilon_{(k)(m)(n)}e^i_{(m)}e^j_{(n)}c_{ij}, \end{aligned} \tag{16}$$

representing the tetrad field strengths $\partial_\mu e_\nu^{(m)} - \partial_\nu e_\mu^{(m)}$ contracted with momentum [28, 29].

Plasma particle-in-cell codes integrate the Lorentz force with the Boris step [6, 26], a velocity-space counterpart of kick-drift-kick leapfrog that brackets a magnetic rotation at constant W between two half-step electric kicks. In the tetrad equation of motion, $\tilde{E}_{(m)}$ carries the lapse, shift, and temporal triad gradients that change W, whereas $\tilde{B}_{(k)}$ carries the spatial triad curl and rotates $u_{(m)}$ at fixed W. Both fields depend on momentum, making the update implicit. Each fixed-point iteration forms $u^n_{(m)} = e^i_{(m)}(t^n, \boldsymbol{x}^n)u_i^n$ and $u^{n+1}_{(m)} = e^i_{(m)}(t^{n+1}, \boldsymbol{x}^{n+1})u_i^{n+1}$ and evaluates $\bar{\tilde{E}}_{(m)}$, $\bar{\tilde{B}}_{(k)}$, and $\bar{W}$ at $\bar{\boldsymbol{x}}$ from their mean $\bar{u}_{(m)}$. With $\Delta\hat{t} = \bar{\alpha}\Delta t$ and $\boldsymbol{u}_{\mathrm{tet}} = (u_{(1)}, u_{(2)}, u_{(3)})$, momentum and position advance through the Boris rotation,

$$\begin{aligned} \boldsymbol{u}^- &= \boldsymbol{u}^n_{\mathrm{tet}} + \frac{\Delta\hat{t}}{2}\bar{\tilde{\boldsymbol{E}}}, \\ \boldsymbol{\vartheta} &= \frac{\Delta\hat{t}}{2\bar{W}}\bar{\tilde{\boldsymbol{B}}}, \\ \boldsymbol{u}' &= \boldsymbol{u}^- + \boldsymbol{u}^- \times \boldsymbol{\vartheta}, \\ \boldsymbol{u}^+ &= \boldsymbol{u}^- + \frac{2(\boldsymbol{u}' \times \boldsymbol{\vartheta})}{1 + |\boldsymbol{\vartheta}|^2}, \\ \boldsymbol{u}^{n+1}_{\mathrm{tet}} &= \boldsymbol{u}^+ + \frac{\Delta\hat{t}}{2}\bar{\tilde{\boldsymbol{E}}}, \\ \frac{x^{i,n+1} - x^{i,n}}{\Delta t} &= \frac{\bar{\alpha}\bar{e}^i_{(m)}\bar{u}_{(m)}}{\bar{W}} - \bar{\beta}^i. \end{aligned} \tag{17}$$

After convergence, $u^{n+1}_{(m)}$ converts back to $u_i^{n+1} = e_i^{(m)}(t^{n+1}, \boldsymbol{x}^{n+1})u^{n+1}_{(m)}$.

Free motion in polar coordinates has straight-line solutions, yet a finite step that evaluates unit-vector rotation at the midpoint and projects at the endpoints makes

an error that Cartesian axes avoid. The spatial triad is such a set of unit vectors, fixed by γ_{ij} only up to a local rotation. Along a continuous geodesic, frame-rotation terms in c_{jk} and c^t_k cancel endpoint projections identically. Over a finite step Δt, midpoint evaluation leaves a second-order truncation error set by this local frame choice, the tetrad gauge. Two gauges are implemented here. The symmetric gauge uses the symmetric positive-definite square root of γ_{ij},

$$e^{(m)}_j = \left(\gamma^{1/2}\right)_{mj}, \tag{18}$$

from eigendecomposition, with spatial derivatives solved from the Sylvester equation for $\partial(\gamma^{1/2})$ in the eigenbasis. The Cholesky gauge uses the lower-triangular factor L_{jm} satisfying $\gamma_{ij} = \delta^{mn} L_{im} L_{jn}$,

$$e^{(m)}_j = L_{jm}, \qquad L_{jm} = 0 \text{ for } j < m, \tag{19}$$

equivalent to Gram–Schmidt orthonormalization of the coordinate basis in index order in closed form at lower cost. Reference [21] leaves its tetrad gauge unstated. On its geodesic benchmarks (Sec. IV), the symmetric gauge reproduces all published digits of the invariant errors for analytic spherical Kerr–Schild input. On the reconstructed Cartesian mesh, the Cholesky gauge in (x, y, z) order reproduces all published digits. Permuted coordinate orderings and cross-paired gauges give distinct errors, while any triad construction that coincides with these two frames along the tested orbits remains consistent with the published digits.

B. Metric reconstruction

Geodesic integration on a mesh requires metric components and first spatial derivatives between cell centers with continuous values and gradients across cell faces. The mesh stores α, β^i, and the six covariant components γ_{ij} at cell centers, and the interpolated 3×3 spatial metric is inverted afterward to preserve symmetry.

Along each axis, a four-point cubic Hermite (Catmull–Rom) stencil spans two cell centers on each side of the bracketing interval at fractional offset $s \in [0, 1)$ in units of the cell spacing h. Interpolation and differentiation evaluate the Hermite weights,

$$\begin{aligned} w_{-1}(s) &= \frac{-s + 2s^2 - s^3}{2}, \\ w_0(s) &= \frac{2 - 5s^2 + 3s^3}{2}, \\ w_1(s) &= \frac{s + 4s^2 - 3s^3}{2}, \\ w_2(s) &= \frac{-s^2 + s^3}{2}, \\ w'_a(s) &= \frac{1}{h}\frac{\mathrm{d}w_a}{\mathrm{d}s}. \end{aligned} \tag{20}$$

Tensor products $w_a(s_1) w_b(s_2) w_c(s_3)$ over the 4^3 surrounding cells reconstruct each field, while substituting $w'_a(s_i)$ along axis i gives the exact polynomial derivative ∂_i. The stencil needs at least two ghost cells around every mesh block, and the runs here use more. Each macroparticle interpolates within its owning block, shifting the stencil inward by one cell to extrapolate the same cubic when an iterate nears the outer edge of the allocated cells. For standalone grid tests, the same routine samples an analytic metric on a virtual uniform lattice of cell centers of spacing h.

C. Stress-energy deposition

Einstein's equations require smooth grid stress-energy components whose proper-volume integrals equal the discrete particle sums. Each macroparticle carries a shape function, a localized profile of unit integral that sets its cloud width on the grid [6, 22]. On a uniform lattice of spacing Δx and cell centers $\boldsymbol{x}_q$, tensor products of the first-order (cloud-in-cell) shape function B_1 define eight nonvanishing lattice weights $\tilde{\omega}_{qp}$,

$$\begin{aligned} B_1(\xi) &= \max\bigl(0,\, 1 - |\xi|\bigr), \\ \tilde{\omega}_{qp} &= \prod_{i=1}^{3} B_1\left(\frac{x^i_q - x^i_p}{\Delta x}\right), \end{aligned} \tag{21}$$

which are nonnegative and sum to unity. Dividing the accumulated moments in leaf cell C by its proper spatial volume $\sqrt{\gamma_C}\,\Delta V_C$, with cell-centered determinant γ_C and coordinate volume ΔV_C, yields the ten deposited sources,

$$\begin{aligned} \bigl(E,\, S_i,\, S_{ij}\bigr)_C = {} & \frac{1}{\sqrt{\gamma_C}\,\Delta V_C} \\ & \times \sum_{p=1}^{N} \omega_{Cp}\, m_p \left(W_p,\, u_{i,p},\, \frac{u_{i,p} u_{j,p}}{W_p}\right), \end{aligned} \tag{22}$$

with dimensionless leaf-cell weight ω_{Cp}.

At a coarse-fine mesh boundary, evaluating B_1 separately on each cell with its own width Δx_C pairs unequal cloud widths on the two sides. For a straddling particle, those weights fail to sum to unity, so mesh integrals differ from the particle sums. A single representation level ℓ_p for each macroparticle p restores unit total weight [21, 30]. Its cloud lives on a virtual uniform lattice of spacing $\Delta x(\ell_p)$, producing eight unit-sum weights $\tilde{\omega}_{qp}$ on virtual cells q of volume ΔV_q (Fig. 2). Those virtual weights map onto leaf cells C of level ℓ_C and volume ΔV_C through

$$\omega_{Cp} = \begin{cases} \displaystyle\sum_{q \subset C} \tilde{\omega}_{qp}, & \ell_C \le \ell_p, \\ \tilde{\omega}_{qp}\, \dfrac{\Delta V_C}{\Delta V_q}, & C \subset q,\ \ell_C > \ell_p. \end{cases} \tag{23}$$

Coarser leaf cells ($\ell_C \le \ell_p$) sum enclosed virtual weights, while finer leaf cells ($\ell_C > \ell_p$) share the virtual cell's coordinate density in proportion to volume. With $\ell_{\rm loc}(p)$ the finest level among leaf blocks owning the eight cell centers nearest $\boldsymbol{x}_p$, setting $\ell_p = \ell_{\rm loc}(p)$ places every cloud at the finest local level, where only the $\ell_C \le \ell_p$ branch occurs. This published rule [21, 30] is used for every result in the paper. Because $\sum_C \omega_{Cp} = 1$ for every particle, summing proper-volume-weighted moments over all leaf cells yields the conservation identity,

$$\sum_C \big(E,\, S_i,\, S_{ij}\big)_C \sqrt{\gamma_C}\, \Delta V_C = \sum_{p=1}^{N} m_p \left(W_p,\, u_{i,p},\, \frac{u_{i,p}u_{j,p}}{W_p}\right), \tag{24}$$

to rounding error across arbitrary refinement boundaries. Enclosing blocks are found by geometric search, and cell updates accumulate via concurrent atomic additions.

Each refinement level halves Δx and divides $N_{\rm ppc}$ by eight, leaving fine cells under diffuse matter with far less than one particle on average. An optional level-capped deposition bounds ℓ_p through the level cap,

$$\ell_p = \min\big(\ell_{\rm loc}(p),\, \ell_{\rm cap}\big), \tag{25}$$

using either a fixed or density-adaptive ceiling $\ell_{\rm cap}$, and recovers the published rule as $\ell_{\rm cap} \to \infty$ (Fig. 2). Both branches of the leaf-weight mapping preserve $\sum_C \omega_{Cp} = 1$, so the conservation identity still holds to rounding error. Coarser clouds raise constraint residuals near forming black holes and shift horizon masses by a few tenths of a percent, more on lattices, yet bound refinement on randomly sampled matter (Sec. V A). Every collapse result in Sec. V uses the published rule.

Short-wavelength sampling fluctuations in (E, S_i, S_{ij}) are smoothed after deposition by $n_{\rm f}$ passes of a separable binomial filter with one-dimensional weights $(1/4, 1/2, 1/4)$ [6, 21]. Ghost cells are filled before each pass by copying at equal level and by restriction and prolongation across refinement levels. Along each axis at wavenumber k, $n_{\rm f}$ binomial passes multiply Fourier modes by $\cos^{2n_{\rm f}}(k\Delta x/2)$ and add $n_{\rm f}\Delta x^2/2$ to the second spatial moment of every cloud. They widen every cloud, which acts as a gravitational softening, a short-range smoothing of the gravitational pull over the filter scale [2]. For a self-gravitating equilibrium, this widening makes the deposited source less concentrated than the matter and lowers the proper-volume integral of E slightly across nonuniform $\sqrt{\gamma}$ (Sec. V).

Plasma particle-in-cell codes cancel this second-moment widening by following the $n_{\rm f}$ binomial passes with one separable compensation pass [6] of one-dimensional weights

$$\big(-c,\, 1+2c,\, -c\big), \qquad c = \frac{n_{\rm f}}{4}. \tag{26}$$

With $s = \sin^2(k\Delta x/2)$, the combined one-dimensional wavenumber response is

$$F(k) = (1-s)^{n_{\rm f}}(1+n_{\rm f}s) = 1 - \mathcal{O}\big((k\Delta x)^4\big). \tag{27}$$

This response equals unity to fourth order in $k\Delta x$, falls monotonically to zero at the Nyquist wavenumber $k = \pi/\Delta x$, cancels the second moment, and carries small negative sidelobes. Cluster evolutions and the randomly sampled prolate collapse in Sec. V apply four binomial passes plus this compensation pass. Spherical dust collapse on a lattice retains one plain binomial pass [21]. With uniform $\sqrt{\gamma}$, both stencils conserve total weight on a single level, whereas mesh-level filtering across coarse-fine ghost zones is nonconservative once particles cross refinement boundaries (Sec. IV).

D. Spacetime evolution

Gravitational collapse and black hole mergers require a hyperbolic spacetime formulation that propagates curvature waves and damps constraint violations. We adopt AthenaK's solver [35, 36] for the Z4c formulation [31, 34, 41] in moving-puncture gauge, the standard coordinate choice for evolving black holes on a fixed Cartesian mesh without excising grid cells [32, 33, 42, 43]. With $K = \gamma^{ij}K_{ij}$, Z4c evolves the conformal variables

$$\chi = \gamma^{-1/3}, \qquad \tilde{\gamma}_{ij} = \chi\gamma_{ij},$$
$$\tilde{A}_{ij} = \chi\left(K_{ij} - \frac{1}{3}\gamma_{ij}K\right), \qquad \hat{K} = K - 2\Theta, \tag{28}$$

alongside the contracted conformal connection $\tilde{\Gamma}^i$ and constraint-propagation scalar Θ, damping the Hamiltonian and momentum constraint residuals $\mathcal{H}$ and $\mathcal{M}_i$ through κ_1 and κ_2. Moving-puncture gauge combines $1+\log$ lapse slicing [42, 44] with the integrated Gamma-driver shift condition [32, 33, 42] damped at rate η with shift speed parameter μ_S,

$$\partial_t\alpha = \beta^i\partial_i\alpha - 2\alpha\hat{K},$$
$$\partial_t\beta^i = \mu_S\tilde{\Gamma}^i - \eta\beta^i + \beta^j\partial_j\beta^i. \tag{29}$$

Cell-centered finite differences—sixth order on refined meshes and fourth order on uniform meshes—advance the state with Kreiss–Oliger dissipation, fourth-order Runge–Kutta integration, and Sommerfeld radiative outer boundaries [36].

AthenaK's Z4c module, previously coupled solely to magnetohydrodynamics [36], is extended here to receive the particle-deposited sources (E, S_i, S_{ij}) in the evolution equations and constraint residuals $\mathcal{H}$ and $\mathcal{M}_i$. An optional parabolic relaxation of the Hamiltonian constraint [21, 45, 46] adds diffusive damping to the conformal factor evolution,

$$\partial_t\chi = (\partial_t\chi)_{\rm Z4c} + \frac{c_H}{2}\chi^{-1/4}\mathcal{H},$$
$$\mathcal{H} = {}^{(3)}R + \frac{2}{3}K^2 - \tilde{A}_{ij}\tilde{A}^{ij} - 16\pi E, \tag{30}$$

where $(\partial_t\chi)_{\rm Z4c}$ is the standard Z4c evolution term and ${}^{(3)}R$ is the Ricci scalar of γ_{ij}. Here the Hamiltonian-constraint relaxation length c_H sums a constant and a

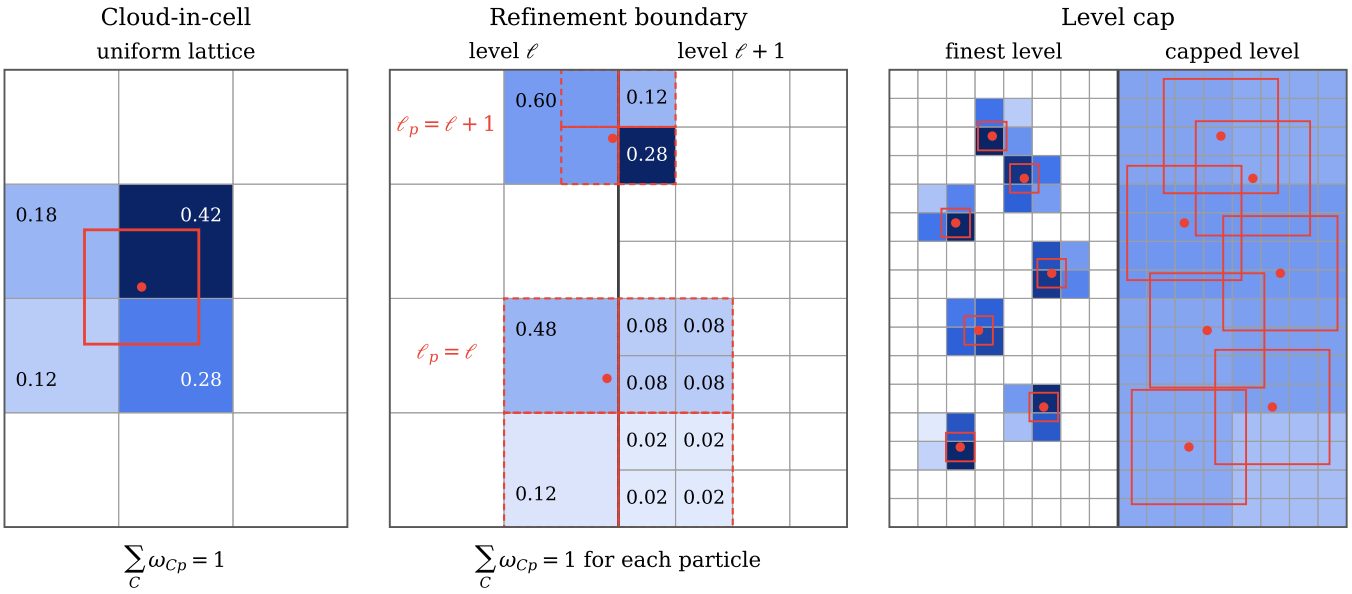


FIG. 2. Single-level stress-energy deposition preserves unit particle weight across refinement boundaries, shown in two dimensions where each refinement level divides particles per cell by four (eight in three dimensions). Left: a cloud-in-cell profile (red square) shares one macroparticle among four uniform cells. Middle: each cloud lives on the lattice of one level ℓ_p (dashed virtual cells). Virtual weights over a coarser leaf cell sum ($\ell_p = \ell+1$), while those over finer leaf cells divide uniformly ($\ell_p = \ell$). Right: identical particles deposited at the finest level and on a capped coarser level; shading marks deposited density and numbers give leaf-cell weights ω_{Cp}.

term proportional to Δx for stability across refinement levels.

E. Adaptive mesh

Gravitational collapse and black hole mergers concentrate steep curvature gradients inside volumes far smaller than the ambient cluster or wave zone. AthenaK uses block-structured octree refinement, splitting cubic mesh blocks recursively into eight half-sized child blocks where finer resolution is required [35, 47]. Uniform and statically refined meshes enter as the limits without dynamic block splitting.

Dynamic refinement tracks the scalar monitor

$$q = \alpha\chi^{-7/4}, \tag{31}$$

which rises sharply in strong-field regions as $\chi^{-7/4}$ grows near black holes [21]. With $\boldsymbol{\iota}_a$ the unit cell shift along axis $a \in \{1,2,3\}$ and $\Delta_a q_i = q_{i+\boldsymbol{\iota}_a} - q_{i-\boldsymbol{\iota}_a}$, each cell i evaluates a Löhner-type second-difference ratio [21, 48], the refinement indicator,

$$\varepsilon_i^2 = \frac{\sum_{a,b=1}^{3}\left(\Delta_a q_{i+\boldsymbol{\iota}_b} - \Delta_a q_{i-\boldsymbol{\iota}_b}\right)^2}{\sum_{a,b=1}^{3}\left(|\Delta_a q_{i+\boldsymbol{\iota}_b}| + |\Delta_a q_{i-\boldsymbol{\iota}_b}| + \epsilon_0 q_{ab,i} + \delta_0\right)^2},$$
$$q_{ab,i} = \sum_{\sigma,\tau=\pm1} \left|q_{i+\sigma\boldsymbol{\iota}_a+\tau\boldsymbol{\iota}_b}\right|. \tag{32}$$

Here ϵ_0 is a dimensionless filter weight and δ_0 is a positive floor. Blocks refine when $\max_i \varepsilon_i$ exceeds an upper threshold and derefine below a lower threshold, while prescribed inner radii enforce a minimum level resolving diffuse matter at the initial particle-lattice scale. Through every regrid, macroparticles keep $(m_p, x_p^i, u_{i,p})$ unchanged without splitting or merging, and metric arrays, block ownership, and deposited sources are rebuilt on the new leaf cells.

F. Numerical methods

End-to-end evolutions from initial equilibria through black hole formation rely on interior particle excision, horizon and mass diagnostics, constraint-satisfying initial data, and portable parallel execution.

Inside a black hole, 1 + log slicing drives α toward zero while signals remain causally trapped behind the horizon. Lapse excision [21, 49] removes particles with $\alpha(t^n, \boldsymbol{x}_p^n) < \alpha_{\rm exc}$, clearing black hole interiors through this lapse criterion alone. Black hole boundaries and irreducible masses are measured on apparent horizons, outermost closed two-surfaces on a slice whose outgoing orthogonal null geodesics have zero expansion. Apparent horizons are located with AthenaK's pseudospectral flow finder [36, 50], searched every few cycles and stopped once the horizon-mass change between iterations falls below a tolerance. As distributed, the finder computes

finite-difference metric derivatives on the active cells of each mesh block and interpolates that derivative array to the trial surface without filling its ghost zones across block faces. Here the metric derivatives on the trial surface are evaluated directly by differentiating the local Lagrange interpolant of the metric, whose ghost cells are filled. Its effect on a black hole of known mass is shown in Sec. V.

An independent check is provided by the Misner–Sharp mass, the gravitational mass inside a coordinate sphere computed from the slice geometry alone,

$$M_{\mathrm{MS}} = \frac{R}{2}\left(1 - \gamma^{ij}\partial_i R\,\partial_j R + R^2 K_T^2\right). \tag{33}$$

Here R is the areal radius of a coordinate sphere of proper area $4\pi R^2$ and K_T is the tangential component of K_{ij} on the sphere. In spherical symmetry, $2M_{\mathrm{MS}}/R = 1$ locates the apparent horizon and M_{MS} equals the black hole mass in vacuum. Values of M_{MS} are evaluated on saved slices independently of the finder. For origin-centered runs, central lapse $\alpha_c(t)$ and central energy density $E_c(t)$ average the eight central leaf cells, defining the central normal-observer proper time,

$$\tau_c(t) = \int_0^t \alpha_c(t')\,\mathrm{d}t'. \tag{34}$$

Coordinate-independent curvature is tracked via the Kretschmann scalar $I = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ of the Riemann tensor $R_{\mu\nu\rho\sigma}$, decomposed through the Gauss, Codazzi, and Ricci equations with Einstein's equations for the two-normal projection [21],

$$\begin{aligned} I &= G_{ijkl}G^{ijkl} - 4C_{ijk}C^{ijk} + 4Z_{ij}Z^{ij}, \\ G_{ijkl} &= {}^{(3)}R_{ijkl} + K_{ik}K_{jl} - K_{il}K_{jk}, \\ C_{ijk} &= D_iK_{jk} - D_jK_{ik}, \\ Z_{ij} &= {}^{(3)}R_{ij} + KK_{ij} - K_{ik}K^k{}_j \\ &\quad - 8\pi\left[S_{ij} - \frac{1}{2}\gamma_{ij}(S - E)\right]. \end{aligned} \tag{35}$$

Here ${}^{(3)}R_{ijkl}$ and ${}^{(3)}R_{ij}$ are the Riemann and Ricci tensors of γ_{ij}, $S = \gamma^{ij}S_{ij}$, and indices are raised with γ^{ij}. This expression is checked against the analytic Schwarzschild profile.

Self-gravitating runs start on conformally flat, time-symmetric slices—surfaces of instantaneous rest with $K_{ij} = 0$, $\beta^i = 0$, and $\gamma_{ij} = \psi^4\delta_{ij}$ for conformal factor ψ—where $\mathcal{M}_i = 0$ identically. Linear superposition of flat-space Poisson potentials u_c for conformal density sources $\tilde{\rho}_c(\boldsymbol{x})$ solves $\mathcal{H} = 0$ via

$$\begin{aligned} \psi(\boldsymbol{x}) &= 1 + \sum_c u_c(\boldsymbol{x}), \\ \nabla^2_{\mathrm{flat}}u_c &= -2\pi\tilde{\rho}_c, \\ E(\boldsymbol{x}) &= \psi(\boldsymbol{x})^{-5}\sum_c \tilde{\rho}_c(\boldsymbol{x}). \end{aligned} \tag{36}$$

Components combine tabulated spherical equilibria, Gaussians, closed-form homogeneous prolate spheroids, and vacuum black hole punctures

$$u_c(\boldsymbol{x}) = \frac{m_c}{2\bar{r}_c} \tag{37}$$

of bare mass m_c at isotropic distance $\bar{r}_c$. With $\sqrt{\gamma} = \psi^6$, equal-mass ensembles of rest-mass density ρ_0 have coordinate number density proportional to $\rho_0\psi^6$. Cold spherical collapse places macroparticles on a mesh-aligned cubic lattice of spacing h, where the published rule [21] assigns rest mass $m_p = \rho_0(\boldsymbol{x}_p)\psi(\boldsymbol{x}_p)^6h^3$ inside the surface. The number of lattice sites inside a sphere differs from its volume in cells by a surface fraction inversely proportional to the cell count across, giving a rest-mass offset at low N. Here a surface correction rescales m_p in the outermost layer of width h by a common factor so $\sum_p m_p$ equals the continuum rest mass M_0. Interior masses stay unchanged, and omitting this rescaling recovers the published rule (Sec. V). Spherical clusters draw particles by randomized stratified sampling in radius and in both position and velocity angles, or by independent Monte Carlo draws in sampling comparisons. The mass-weighted mean of $u_{i,p}$ is subtracted so net cluster momentum vanishes, while prolate spheroids use equal-mass Monte Carlo sampling with reflection and half-turn symmetries.

All mesh and particle kernels run through Kokkos [35, 37] across multicore processors and graphics accelerators, with particle updates taking a small fraction of the spacetime step cost at $N \le 10^4$. Two optional features of Ref. [21] are not implemented here: the auxiliary-field shift condition and position-dependent profiles of η and κ_1 tied to tracked black holes.

IV. TEST PARTICLES

Passive trajectories in a known spacetime isolate geodesic integration, metric reconstruction, deposition, and interior excision from gravitational back-reaction. In this test-particle regime, macroparticles follow timelike geodesics of a fixed or independently evolved vacuum metric while deposited sources (E, S_i, S_{ij}) serve as passive diagnostics. Direct comparison with exact geodesics and published benchmarks [21] tests every particle stage in units of the black hole ADM mass M.

A. Fixed black hole

Bound orbits around a spinning black hole test whether both pushers preserve relativistic integrability and precession over many dynamical periods. Around a Kerr black hole of spin a in Boyer–Lindquist or spherical Kerr–Schild coordinates (t, r, θ, ϕ), timelike geodesics conserve the Killing energy $\mathcal{E}$, axial angular momentum L_z, and

Carter constant Q,

$$\mathcal{E} = -u_t, \qquad L_z = u_\phi,$$
$$Q = u_\theta^2 + \cos^2\theta \left[a^2\left(1-\mathcal{E}^2\right) + \frac{L_z^2}{\sin^2\theta} \right]. \tag{38}$$

In the nonrotating limit ($a \to 0$), spherical symmetry reduces Q to $L^2 - L_z^2$ for total angular momentum L. At spin $a = 0.9\,M$, equatorial eccentric and inclined Boyer–Lindquist orbits stay on their invariant tori over many radial periods (Fig. 3). Both pushers match exact apsidal and nodal precession with bounded invariant errors and convergent fixed-point iterations. Because the Boyer–Lindquist spatial metric is diagonal, both tetrad gauges coincide there, and the midpoint update conserves cyclic L_z to rounding error [27].

A near-capture eccentric orbit probes strong-field accuracy against the benchmarks of Ref. [21]. In Kerr–Schild coordinates, an inclined trajectory ($a = 0.9\,M$, $e = 0.99$) passes close to the event horizon before returning outward (Fig. 4). Because the midpoint update of Sec. III A depends solely on the coordinate metric and its spatial derivatives, its discrete trajectory is uniquely fixed. On the analytic spherical Kerr–Schild metric, the implicit midpoint pusher conserves cyclic L_z to rounding error and matches all published digits of the maximum relative errors in $\mathcal{E}$ and Q [21]. On the Cartesian Kerr–Schild metric reconstructed with the Hermite weights of Sec. III B, where the mesh breaks continuous axisymmetry, it again matches all published digits for $\mathcal{E}$, L_z, and Q [21].

Off-diagonal spatial metric terms in Kerr–Schild coordinates separate the symmetric gauge from the Cholesky gauge in the spacetime Boris pusher. Continuous geodesics are frame independent because triad connections in the tetrad equation of motion cancel endpoint projections identically. Over a finite step Δt, midpoint evaluation of the Boris rotation leaves a gauge-dependent second-order error. On analytic spherical Kerr–Schild input, the symmetric gauge reproduces all published digits of the $\mathcal{E}$, L_z, and Q errors in Ref. [21] (Fig. 4), whereas the (r, θ, ϕ) Cholesky gauge gives distinct values. On the mesh-reconstructed Cartesian metric, the (x, y, z) Cholesky gauge reproduces all published digits, whereas the symmetric gauge and permuted Cholesky orderings give distinct values.

Independent sweeps of Δt and cell spacing h measure temporal and spatial convergence across spinning and nonrotating black hole spacetimes. On analytic metrics, position and invariant errors of both pushers and gauges fall as Δt^2 across the bound Kerr orbits and the $e = 0.99$ passage (Fig. 4). In the nonrotating limit ($a \to 0$), circular, inclined eccentric, and plunging orbits are tested against exact Schwarzschild solutions (Fig. 5). Spherical Kerr–Schild circular motion advances linearly in time and azimuth and integrates to rounding error. Cartesian circular, eccentric, and plunging orbits converge as Δt^2, and multi-block mesh runs at fine Δt match virtual-lattice runs of spacing h to rounding error across block faces. Grid refinement in h reduces metric, position, and invariant errors at third order for both $a = 0.9\,M$ and $a \to 0$ (Fig. 5).

Analytic Cartesian trajectories separate spatial interpolation error from Cartesian time-stepping error on the $e = 0.99$ passage. Reference [21] attributes the mesh error of both pushers on this orbit to spacetime interpolation. Time-step sweeps on the analytic Cartesian metric alongside the Hermite mesh at fixed h isolate each contribution (Fig. 4). Spatial interpolation sets the $(\mathcal{E}, L_z, Q)$ errors on the mesh, which exceed analytic Cartesian values at the published step size and flatten onto an interpolation floor as Δt shrinks. Conversely, the final-position error on the analytic Cartesian metric without interpolation matches the Hermite mesh closely at the published step size, and both fall as Δt^2. Direct differences between mesh and analytic-Cartesian final states at equal Δt confirm that spatial interpolation shifts the position far less than Cartesian time-stepping down to the finest steps tested.

Orbiting clouds of macroparticles test phase mixing and invariant conservation on a refined mesh. Gaussian clusters of $N = 10^2$, 10^3, and 10^4 bound test particles start around an equatorial circular orbit for $a = 0.9\,M$ and $a \to 0$. Each cluster evolves on both the analytic Cartesian metric and a statically refined octree mesh. Differential orbital shear stretches each cloud into an arc, a trailing spiral, and an azimuthally mixed ring traced by E [2] (Fig. 6). Single-particle $(\mathcal{E}, L_z, Q)$ errors stay bounded without secular drift, independently of N. On the mesh, both pushers give matching invariant errors that decrease at third order with h, and both follow the analytic phase-mixing histories closely.

Repeated octree-interface crossings during phase mixing and infall test multi-level source conservation before and after filtering. Every macroparticle deposits (E, S_i, S_{ij}) at the finest local level (Sec. III C). Before filtering, the conservation identity holds to rounding error across all level crossings and excisions whenever clouds sit at least half a cell inside the outer boundary (Fig. 6). For the few wide-orbit particles within half a cell of the outer boundary, the cloud weight outside the mesh accounts to rounding error for the difference. After the filter on the mesh, the integrals match the particle sums before level crossings begin and differ slightly once particles cross refinement boundaries, reflecting the nonconservation of the mesh filter across levels.

Clusters near the capture barrier test lapse excision against exact geodesics and measure sampling-noise scaling with N. Plunging-family clouds ($N = 10^4$) at $a = 0.9\,M$ and $a \to 0$ send many macroparticles across the event horizon, where exact Kerr effective potentials predict arrival times at $\alpha_{\rm exc}$. With both pushers on the mesh, lapse excision matches the predicted removal curve up to a few marginal orbits near the capture threshold perturbed by interpolation (Fig. 6). Active, excised, and escaped rest masses sum to the initial cluster mass to rounding error. Across independent bound clusters of

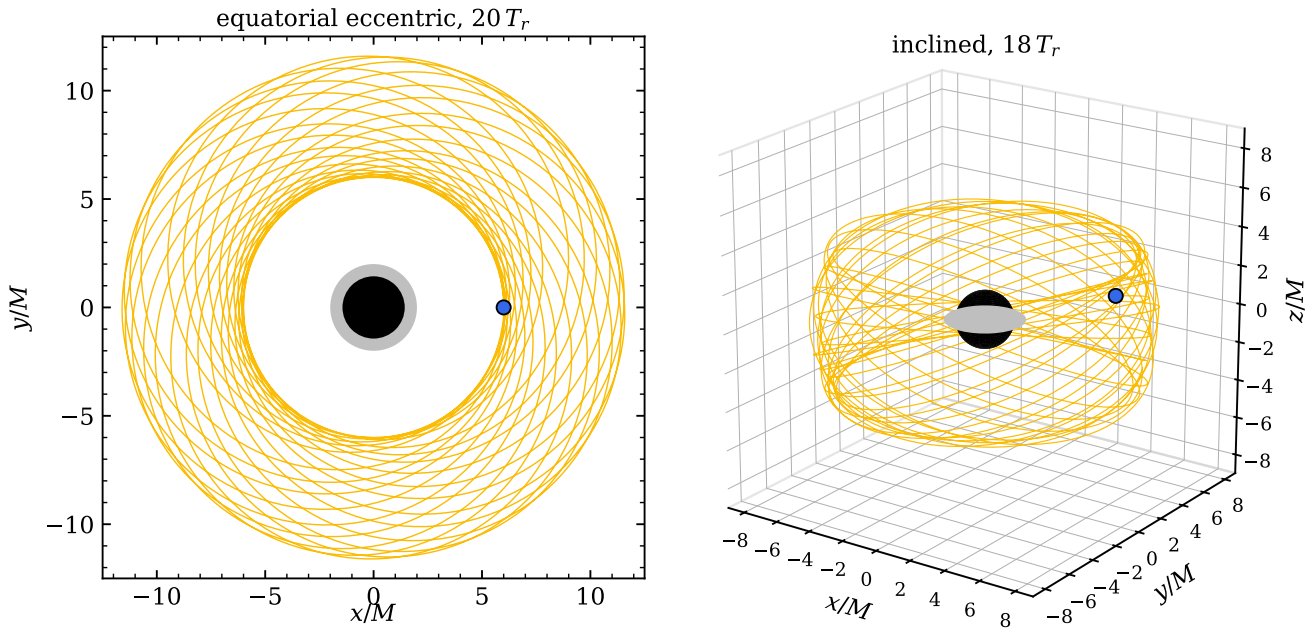


FIG. 3. Bound test-particle trajectories around a Kerr black hole ($a = 0.9\,M$) precess coherently and stay confined to their invariant tori. Left: equatorial eccentric orbit; right: inclined three-dimensional orbit. Solid yellow curves show the spacetime Boris pusher in the symmetric gauge on the analytic Boyer–Lindquist metric from initial blue markers outside the black event horizon and gray ergosphere.

$N = 10^2$, 10^3, and 10^4 macroparticles at fixed total mass, pairwise root-mean-square differences in filtered E fall as $N^{-1/2}$.

B. Evolving black holes

Dynamical black hole evolutions in moving-puncture gauge mix physical geodesic motion with coordinate drift. A nonrotating vacuum puncture of initial mass M evolves on a statically refined Cartesian mesh under Z4c (Sec. III D) from time-symmetric data with a precollapsed lapse that starts small near the puncture. Zero-mass tracers advance on the time-averaged Hermite-reconstructed metric with the implicit midpoint pusher. As $1 + \log$ slicing and the Gamma-driver shift relax the slice toward a stationary trumpet [32, 33, 51], coordinates drift outward and $\alpha W - \beta^i u_i$ varies during the gauge transient. Comparison with exact Schwarzschild geodesics therefore uses gauge-independent worldline observables: orbital azimuth, the areal radius R of the coordinate sphere through the particle, and proper time

$$\tau(t) = \int_0^t \frac{\alpha}{W}\,\mathrm{d}t'. \tag{39}$$

Radially infalling and plunging tracers test strong-field worldlines through the evolving apparent horizon to interior excision. A radial infaller from rest and an inspiraling plunger cross refinement boundaries smoothly, enter the apparent horizon, and undergo lapse excision once the interior lapse drops below α_{exc} (Fig. 7). Along radial infall, $R(\tau)$ tracks the exact Schwarzschild trajectory closely outside the horizon, departing only within a few cells of the puncture where steep gradients limit interpolation of R. Along the slower plunge, $R(\tau)$ and azimuth accumulate a small advance relative to the exact geodesic of mass M. Exact geodesic integration from the post-transient state with the slice Misner–Sharp mass M_{MS} removes nearly that entire offset (Fig. 7).

Circular tracers evolved over many periods test whether their small radial oscillations reflect pusher error or the black hole mass in the simulation. Equatorial and inclined circular tracers keep planar orbits and conserved angular momentum L while their coordinate radii drift outward and $R(\tau)$ oscillates slightly inside its initial value. Saved slices show that the exterior settles into a stationary Schwarzschild slice whose Misner–Sharp mass M_{MS} slightly exceeds M at this resolution. With the angular momentum of mass M, each tracer orbits that slightly heavier numerical black hole from apocenter on a mildly eccentric geodesic matching $R(\tau)$. In Schwarzschild geometry, R, L, and proper radial acceleration imply an enclosed mass

$$M_{\mathrm{eff}} = \frac{L^2/R - R^2\,\mathrm{d}^2R/\mathrm{d}\tau^2}{1 + 3L^2/R^2}, \tag{40}$$

which agrees across all orbital radii and inclinations with M_{MS} of the slice (Fig. 7).

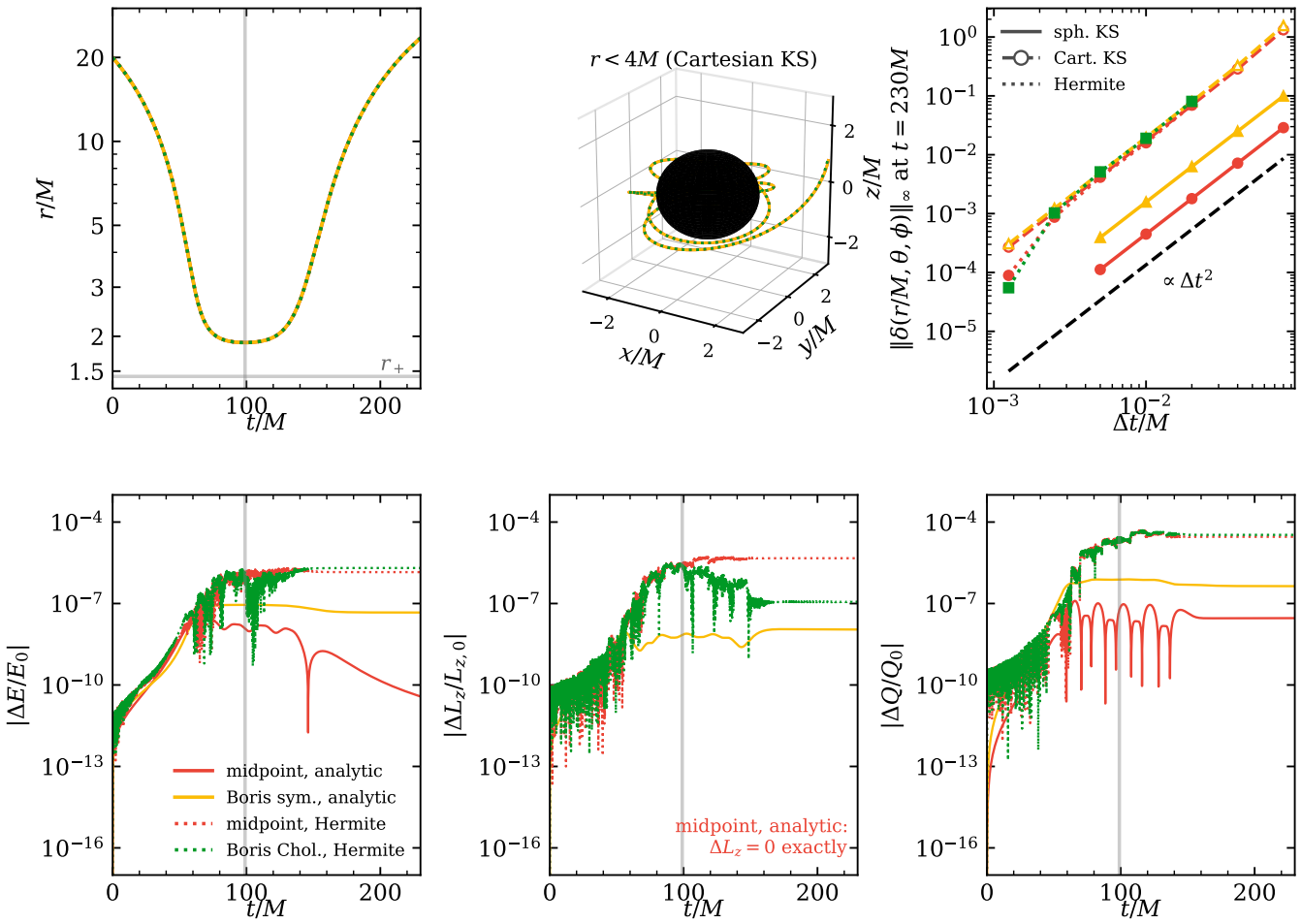


FIG. 4. A near-capture Kerr passage ($a = 0.9\,M$, $e = 0.99$) conserves $(\mathcal{E}, L_z, Q)$ with both pushers and converges as Δt^2. Top: radius against time, trajectory near the black hole, and position error against time step with a second-order guide (black dashed). Bottom: relative errors of $\mathcal{E}$, L_z, and Q against time across periapsis (gray line). Red marks the implicit midpoint pusher, yellow the spacetime Boris pusher in the symmetric gauge, and green the spacetime Boris pusher in the Cholesky gauge. Solid, dashed, and dotted curves mark analytic spherical Kerr–Schild, analytic Cartesian, and mesh-reconstructed metrics.

A head-on collision of two equal-mass black holes starting from rest extends these tracer tests to a dynamical merger with adaptive mesh refinement following both punctures. Axisymmetry about the collision axis makes each particle's angular momentum about that axis the one exact geodesic invariant of the spacetime. Orbiting tracers conserve that angular momentum closely through merger within the axisymmetry truncation error of the Cartesian mesh, while a symmetry-plane tracer from rest holds zero angular momentum to rounding error until excision. Once the punctures meet, the mass of the common apparent horizon exceeds the total mass slightly, measuring the error of the spacetime evolution at this resolution.

V. SELF-GRAVITATING MATTER

Coupled Einstein-Vlasov dynamics begins once macroparticles carry finite rest mass m_p and feed their deposited sources (E, S_i, S_{ij}) back into the spacetime evolution. Collective gravitational wells then form, oscillate, disperse, or collapse into black holes from the matter distribution itself, closing the particle-mesh cycle of Sec. III. Whereas Ref. [21] evolves self-gravitating benchmarks with 10^6 or more macroparticles on large machines, every run here uses at most ten thousand macroparticles ($N \leq 10^4$) on one graphics card. Quantities independent of N, such as continuum equilibrium sequences and analytic collapse bounds, test the coupled solver at full precision. Phenomena sensitive to discrete sampling—initial lattices, multi-pass filtering, refinement under diffuse matter, and cluster holding—isolate what changes at $N \leq 10^4$, with scenario parameters

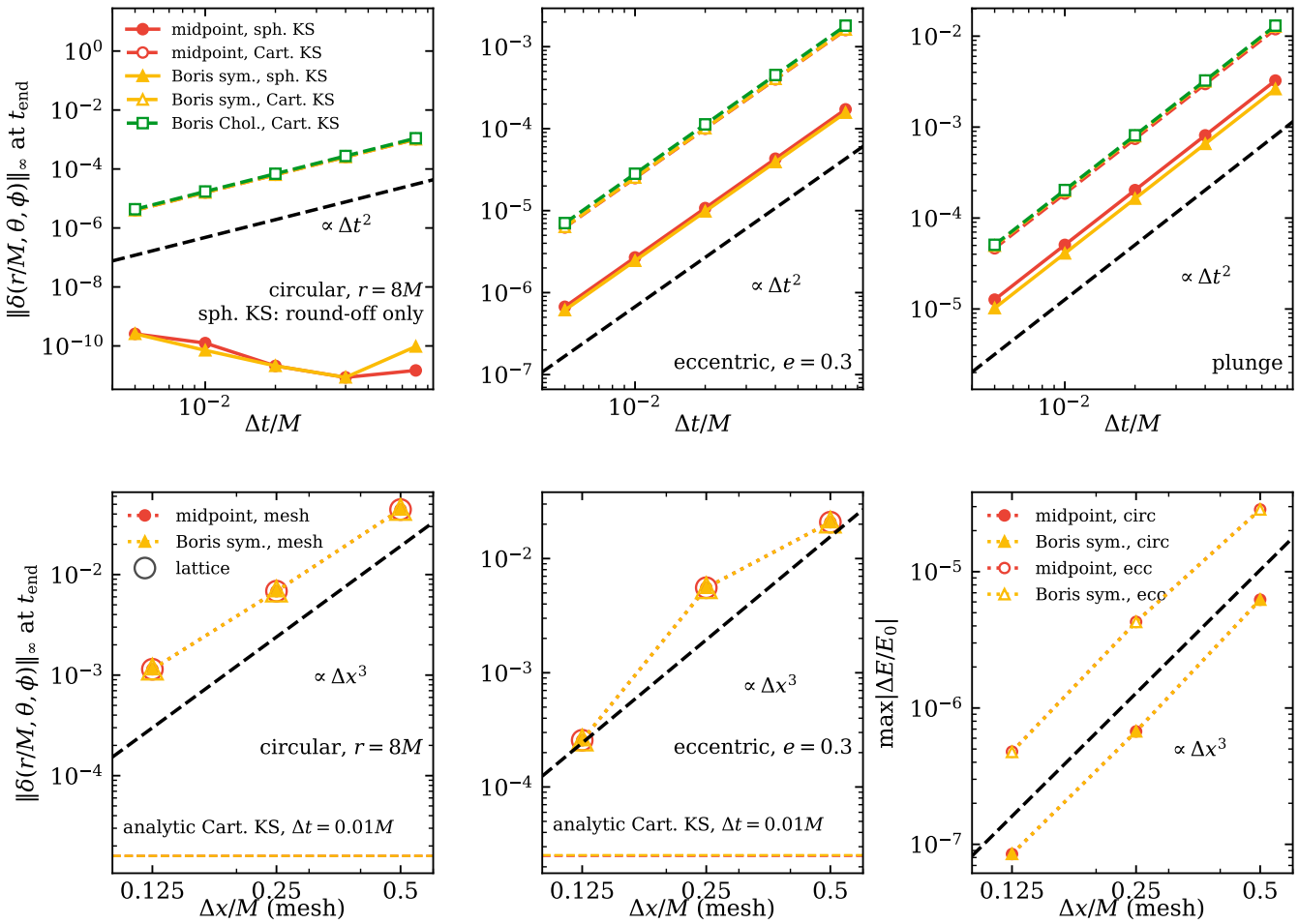


FIG. 5. In the nonrotating limit ($a \to 0$), trajectories converge at second order in Δt on analytic metrics and at third order in cell spacing h under cubic Hermite reconstruction. Top: position error versus Δt for circular, eccentric, and plunging orbits; bottom: position and $\mathcal{E}$ errors versus h on the mesh (dotted) and virtual lattice (open markers) with power-law guides (black dashed). Curves show the implicit midpoint pusher (red) and the spacetime Boris pusher in the symmetric (yellow) and Cholesky (green) gauges on analytic spherical (solid) and Cartesian (dashed) metrics.

listed in the appendix.

A. Dust collapse

Spherical collapse of cold, pressureless matter from rest tests the coupled solver from smooth initial curvature through black hole formation against an exact solution [38]. A homogeneous dust ball of initial areal radius $R_0 = 5\,M$, ADM mass M, and uniform density $E_0 = \rho_0 = 3M/(4\pi R_0^3)$ starts at rest on a conformally flat slice [21, 52, 53]. At $t = 0$, momenta $u_{i,p}$, extrinsic curvature K_{ij}, and shift β^i vanish, $\alpha = 1$, and interior and exterior ψ join smoothly at isotropic radius $\bar{r}_0$. Nearly 10^4 macroparticles occupy a cubic lattice of spacing h inside $\bar{r}_0$ and evolve with dynamic refinement, one plain binomial filter pass, and lapse excision. Along free-falling worldlines parameterized by $\varphi \in [0, \pi)$, shell areal radius R, central proper time τ_c, and interior density E follow the cycloid [38],

$$R(\varphi) = \frac{R_0}{2}(1 + \cos\varphi), \qquad \tau_c(\varphi) = \sqrt{\frac{R_0^3}{8M}}\,(\varphi + \sin\varphi),$$
$$E(\varphi) = E_0\left(\frac{R_0}{R(\varphi)}\right)^3. \tag{41}$$

Under 1+log slicing without shift advection from unit initial lapse, local integration inside the homogeneous core gives an analytic lower bound on α_c, the lapse bound [53],

$$\alpha_{\rm bound}(\tau_c) = 1 + 6\ln\left(\frac{R(\tau_c)}{R_0}\right). \tag{42}$$

Until a coordinate slicing wave launched at the surface density jump reaches the origin, causal isolation keeps $\alpha_c(\tau_c)$ on the lapse bound while the dust stays in free fall.

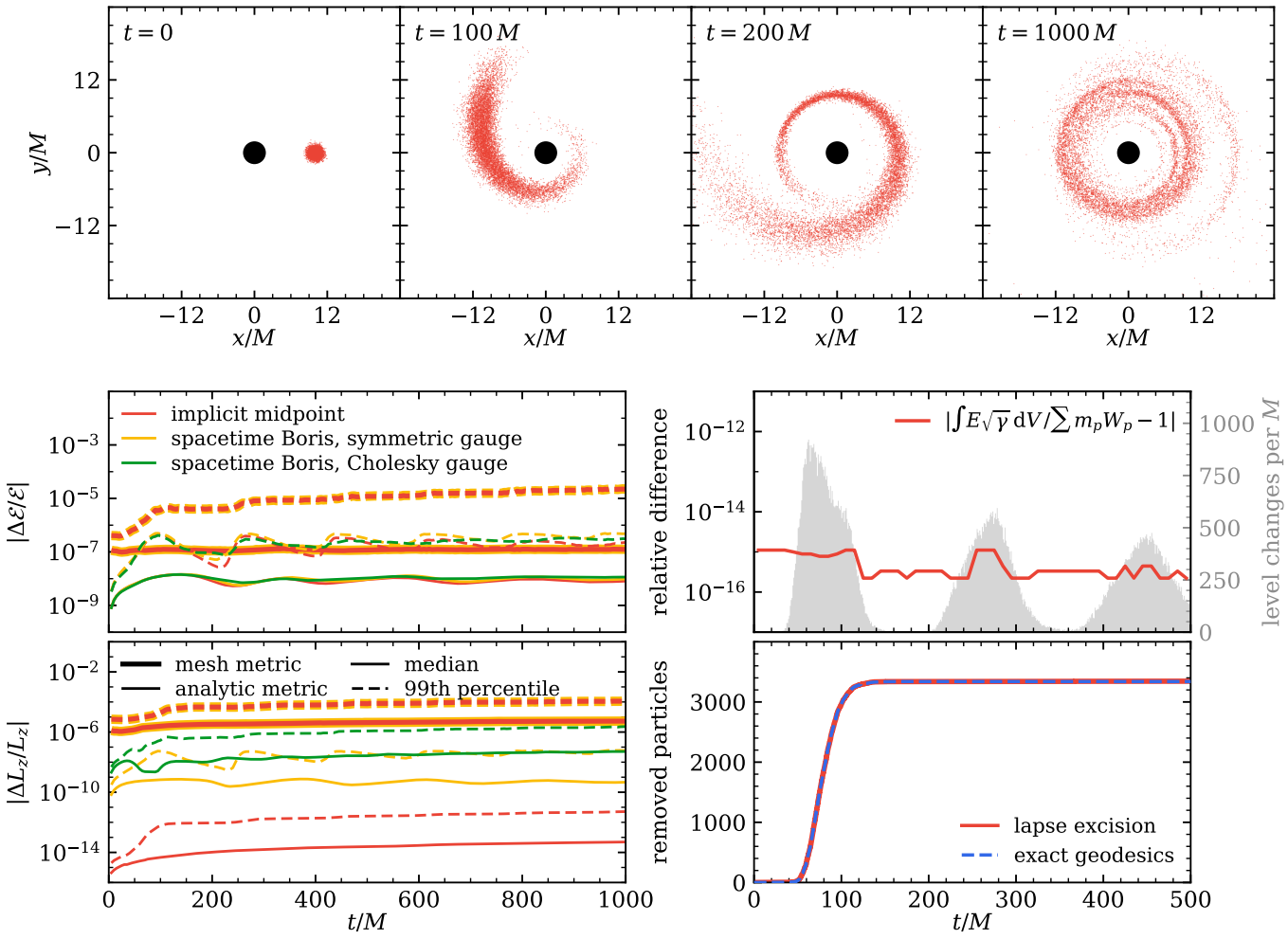


FIG. 6. Clusters of 10^4 test particles around a black hole of spin $a = 0.9\,M$. Top: a cluster started on a circular orbit phase-mixes into a ring (orbital plane, mesh metric; black disk: the black hole). Bottom left: median (solid) and 99th percentile (dashed) of the relative errors of $\mathcal{E}$ and L_z. Color marks the pusher, thick curves the mesh metric (yellow edges: the spacetime Boris run under the implicit midpoint run), and thin curves the analytic metric. Bottom right, for a plunging-family cluster: the conservation identity before the filter (red) against the rate of refinement-level changes (gray, right axis). Lower panel: particles removed by lapse excision (red) against exact geodesics (blue dashed).

Central infall and black hole formation are expected to match continuum Oppenheimer–Snyder collapse despite the discrete particle representation. In the simulation (Fig. 8, top left), $\alpha_c(\tau_c)$ tracks the lapse bound closely during causal isolation and departs smoothly once the surface coordinate wave arrives, matching Ref. [21] and continuum runs [53]. Before dynamic refinement, the central density $E_c(\tau_c)$ likewise follows the cycloid closely (Fig. 8, top right, red curve). Once curvature refinement subdivides the central blocks, fine cells outnumber nearby particles and leave part of the central probe empty, dropping E_c below the cycloid until infall concentrates particles inward. As strong-field curvature grows, $1 + \log$ slicing drives α_c toward zero and an apparent horizon forms around the matter (Fig. 8, bottom left). Shortly afterward, interior α drops below $\alpha_{\rm exc}$ and lapse excision removes every macroparticle, while the Hamiltonian constraint norm $\|\mathcal{H}\|$ peaks and damps across the vacuum ringdown (Fig. 8, bottom right).

At $N \lesssim 10^4$, discrete lattice initialization must match the particle rest-mass sum to the continuum rest mass M_0 of the initial slice. Under the published rule [21], each site inside $\bar{r}_0$ carries rest mass $m_p = \rho_0 \psi(\boldsymbol{x}_p)^6 h^3$, the local proper rest-mass density times the cell volume. That rule was first assumed to yield a black hole of mass M here, as at $N \sim 10^6$ where surface count offsets vanish. At $N \lesssim 10^4$, however, more lattice sites lie inside the sphere than its volume in cells, making $\sum_p m_p$ heavier than M_0 assumed by ψ. Runs with published masses kept the interior density exact—so α_c and E_c still matched the analytic collapse—yet left an apparent-horizon mass $M_{\rm AH}$ visibly above M (Fig. 8, orange curve). Radial profiles of the slice Misner–Sharp mass $M_{\rm MS}$ traced this excess to a mass shell leaving the boundary at $t = 0$ at light

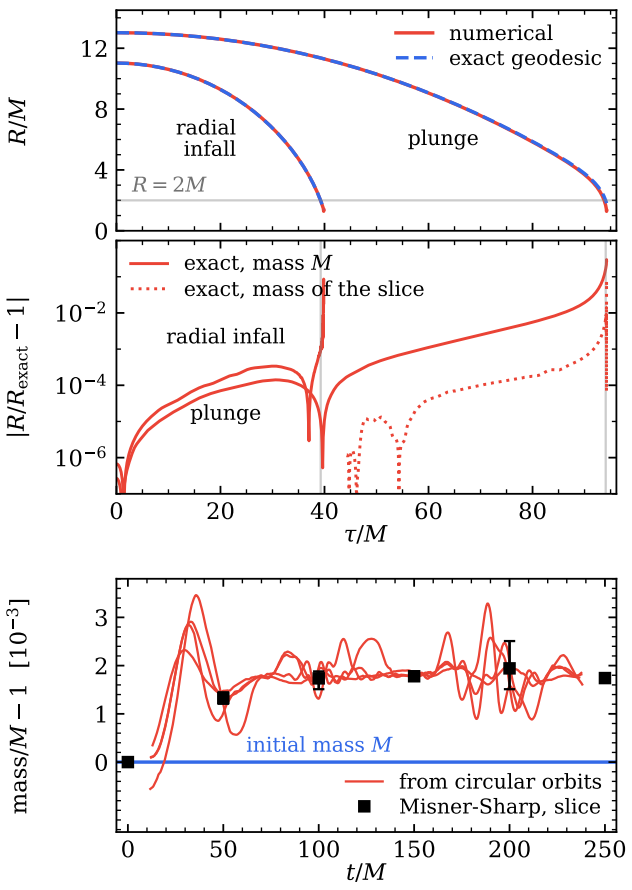


FIG. 7. Test particles around an evolved puncture of mass M. Top: areal radius against proper time for infalling tracers (red) and exact geodesics (blue). Middle: relative difference with mass M (solid) or slice mass $M_{\rm MS}$ (dotted); gray: $R = 2M$. Bottom: circular-orbit mass (red) and $M_{\rm MS}$ (black).

speed, outside which $M_{\rm MS}$ equaled M. The surface correction of Sec. III F rescales m_p in the outermost lattice layer of width h so $\sum_p m_p = M_0$ with interior masses unchanged. With corrected masses, $M_{\rm AH}$ at formation matches M to within the accuracy of the mass measure and drifts downward only slightly afterward (Fig. 8, red curve).

Reliable horizon masses after collapse require checking the apparent-horizon finder on a vacuum black hole of known mass M without matter or excision. In initial collapse tests, AthenaK's distributed pseudospectral flow finder [35, 36, 50] returned horizon masses visibly below M. A first reading, that the flow iteration stopped too early, was ruled out on an evolved vacuum puncture of mass M (Sec. IV B) when a much tighter stopping rule left the deficit unchanged. Across three cell spacings on that puncture, AthenaK's distributed finder reported masses well below M, with a larger deficit at finer resolution (Fig. 9, top panel, solid curves). Independent evaluations of the slice Misner–Sharp mass $M_{\rm MS}$—the enclosed mass computed from slice geometry alone (Sec. III F)—converged rapidly toward M on the same slices (Fig. 9, open circles). That check showed that the evolved spacetime carried the mass M to the accuracy plotted and that the deficit belonged to the finder. Reading the finder then showed that it interpolates metric derivatives stored on active cells, whose ghost zones are left unfilled across block faces (Secs. III F and VI). The deficit is largest on the finest mesh, where the blocks are narrowest and more block faces cut the origin-centered horizon. Direct differentiation of the ghost-filled Lagrange metric interpolant on the trial surface eliminates those block-face errors and restores spherical zero-expansion horizons. The corrected apparent-horizon mass $M_{\rm AH}$ then matches $M_{\rm MS}$ closely on saved slices and in live runs (Fig. 9, dashed curve and diamonds).

Level-capped deposition (Sec. III C) applies the level cap with $\ell_{\rm cap}$ set by local particle density at $N \le 10^4$. On the surface-corrected lattice, an unlimited cap removes the $\|\mathcal{H}\|$ jump at the first refinement (Fig. 8; Table II). The central-density reading $E_c(\tau_c)$ then sits above the cycloid. Innermost particle radii stay identical across the runs. That excess arises because the contracting lattice beats against a cloud lattice of nearly equal spacing at the center. Stored particles from the uncapped run reproduce these E_c readings when redeposited onto lattice-level clouds. Near the forming black hole, the cap raises $\|\mathcal{H}\|$ and drives $M_{\rm AH}$ above M after excision. A one-level cap raises E_c further, elevates $\|\mathcal{H}\|$, and shifts $M_{\rm AH}$ to about one percent below M (Fig. 8; Table II).

On a random sample of the same ball, full-depth refinement under the published rule follows sampling noise into runaway block growth before an apparent horizon forms, with plain or compensated filtering. Capped clouds complete those runs with $M_{\rm AH}$ at M to a few tenths of a percent (Table II). At equal settings one level shallower with compensated filtering, both rules complete on the random sample. Capped clouds still raise $\|\mathcal{H}\|$ after black hole formation and shift $M_{\rm AH}$ lower by a few tenths of a percent (Table II). A first explanation, that the lattice beat alone caused this disturbance, was tested and refuted by that equal-settings run. Coarser clouds raise constraint residuals near a forming black hole and shift $M_{\rm AH}$ by a few tenths of a percent, more on a lattice, yet bound deep refinement on random samples. The published rule is more accurate where refinement stays bounded and is used for every collapse result in the paper.

B. Prolate collapse

Cold prolate collapse from rest tests the coupled solver in three dimensions beyond spherical symmetry. Transverse infall focuses collisionless matter onto the symmetry axis into a caustic before any apparent horizon forms. Early axisymmetric simulations found rapidly growing curvature ahead of any apparent horizon [54]. Analytic examples showed that spatial slices of a black hole space-

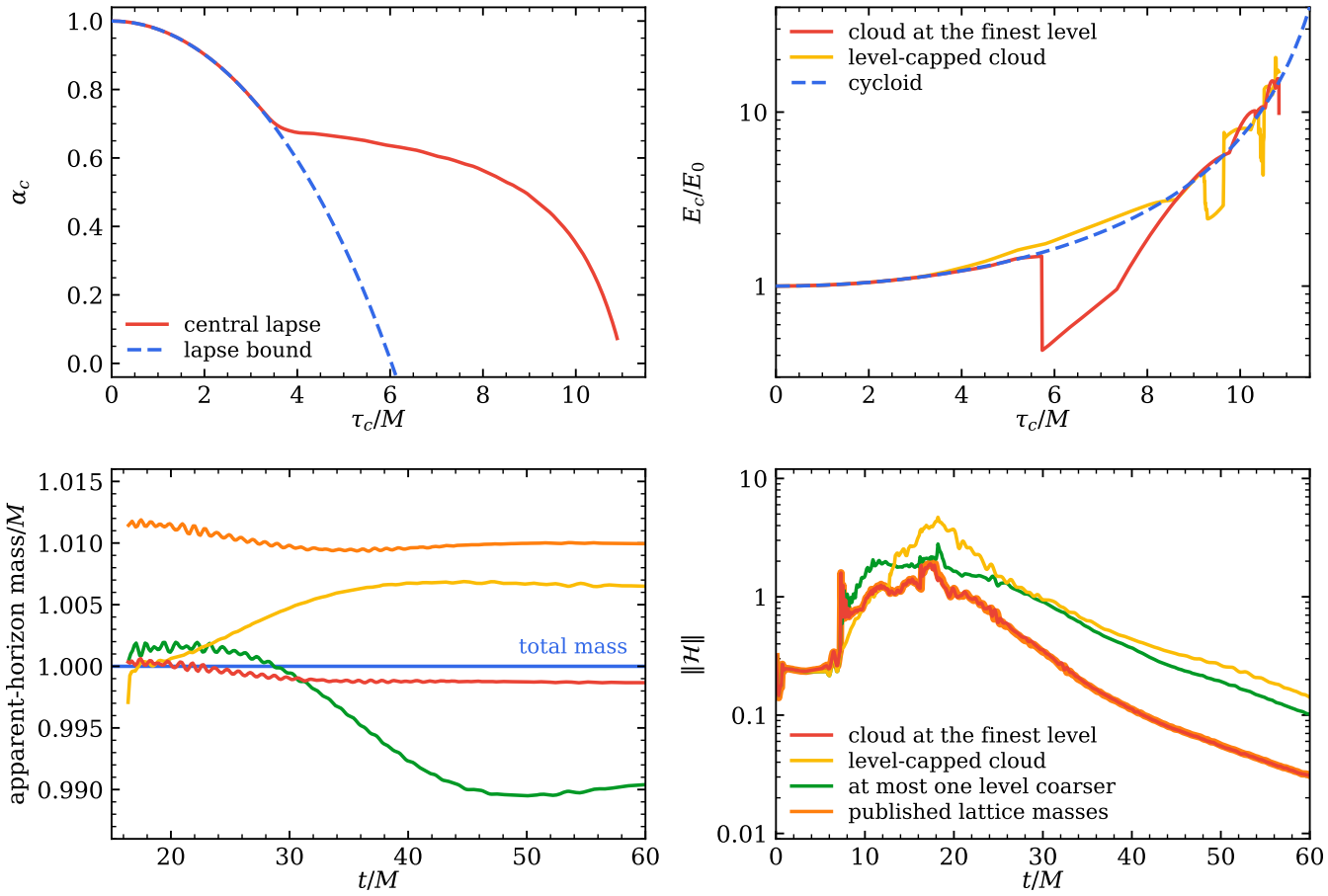


FIG. 8. Gravitational collapse of a pressureless dust ball ($R_0 = 5\,M$, nearly 10^4 macroparticles) to a black hole on an adaptive mesh. Top left: central lapse α_c versus central proper time τ_c (red) and the analytic lapse bound (blue dashed). Top right: central energy density E_c/E_0 and the cycloid (blue dashed) before central excision. Bottom: apparent-horizon mass with total mass M (blue), and Hamiltonian constraint norm $\|\mathcal{H}\|$. Colors mark finest-level clouds with surface-corrected (red) or published (orange) lattice masses, and level-capped clouds with unlimited (yellow) or one-level (green) coarsening.

time can miss trapped surfaces [55]. Three-dimensional particle runs later found resolution-dependent curvature peaks inside the spindle for this model [56]. Generalized-harmonic evolutions then followed collapse past the caustic to black hole formation [19]. Particle-in-cell Z4c runs likewise formed an apparent horizon after the caustic [21]. A homogeneous spheroid of mass M and eccentricity $e = 0.9$ starts from rest with $N = 10^4$ symmetric equal-mass macroparticles (Appendix B) on an adaptive mesh (Table I).

Radial infall contracts the spheroid into a thin spindle (Fig. 10, top row, at saved snapshots nearest the published times [21]). At $N = 10^4$ this spindle is an axial chain of separate condensations (Fig. 10, third panel). Equatorial reflection symmetry of the sample places two mirror-image maxima of the Kretschmann scalar I (Sec. III F) inside the matter on the axis, first near the ends [21]. Those maxima move inward in steps as peak I jumps between condensations (Fig. 10, bottom left). Peak I depends on cell spacing [21], with an upward jump in Fig. 10 when the finest mesh levels appear. With grid curvature finite throughout focusing, the run continues through the stage of largest curvature. An apparent horizon first forms at the time shown in Ref. [21], with spindle tips extending past its poles (Fig. 10). Lapse excision below $\alpha_{\rm exc}$ then removes all 10^4 macroparticles as those tips fall into the black hole.

The apparent-horizon mass $M_{\rm AH}$ (Sec. III F) peaks a few percent below M after polar infall (Fig. 10, bottom right). That apparent-horizon mass continues to fall at the end of the run. Reference [21] instead reports agreement with M at higher particle number and finer resolution. The slice Misner–Sharp mass $M_{\rm MS}$ (Sec. III F) on the final slice agrees with the finder at the apparent horizon. The same profile $M_{\rm MS}$ recovers the total mass M at large radius. The remaining few percent of M after lapse excision reside in gravitational fields outside the black hole. That enclosed geometric mass combines gravitational radiation and constraint violation in one total. The Hamiltonian constraint norm $\|\mathcal{H}\|$ peaks at largest curva-

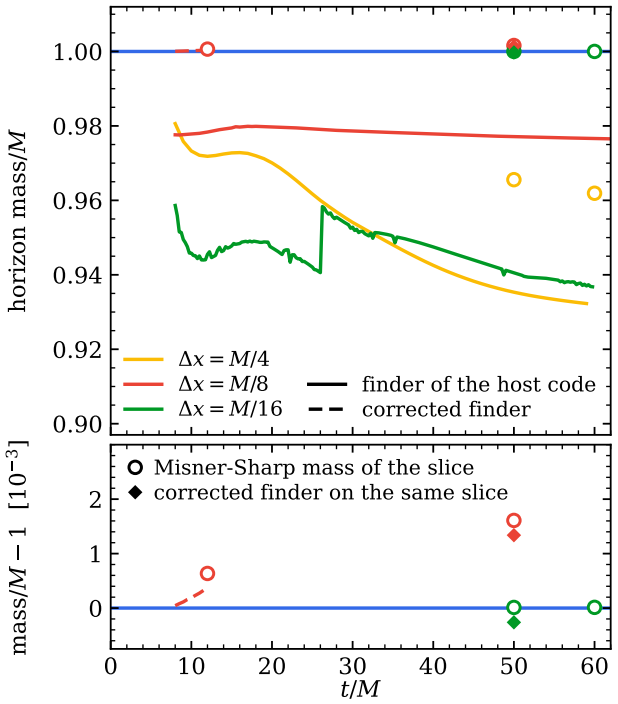


FIG. 9. Apparent-horizon mass of a vacuum black hole of mass M (blue) at three cell spacings (colors). Solid curves show the host code's finder, and the dashed curve differentiates the metric interpolant. Circles show the slice Misner–Sharp mass and diamonds the corrected finder on the same slices (enlarged below).

ture several times above its lattice dust peak (Sec. V A). A spherical control with a random sample and the same filter loses only a few tenths of a percent of M, so random sampling alone does not explain this deficit. How the deficit of the prolate collapse splits between particle number and cell size at its caustic stage is not determined. Dynamic refinement (Sec. III E) with halved finest cells exceeds the block memory of one graphics card, which sets our finest grid.

On a randomly drawn dust ball at full refinement depth, discrete sampling noise under the published rule drives unbounded block growth with plain or compensated filtering. Level-capped clouds keep that full-depth refinement bounded through black hole formation (Sec. V A). Four binomial passes with compensation (Sec. III C) keep the block count bounded throughout the prolate collapse.

C. Relativistic clusters

Warm collisionless star clusters in equilibrium balance collective gravity through velocity dispersion. Their evolution tests phase-space conservation alongside general-relativistic radial stability [1, 10–12]. As in Shapiro and Teukolsky [1, 12] and Ref. [21], static, spherically symmetric, isotropic monoenergetic clusters populate one Killing energy $\mathcal{E} = \alpha W = \alpha_{\rm s}$. Here $\alpha_{\rm s}$ is the boundary lapse where particles turn around ($W = 1$). Radial integration of the Einstein-Vlasov equilibrium relations—the structure equations—yields a one-parameter sequence labeled by $y_c \in (0, 1)$, with gravitational binding energy E_b,

$$y_c = \left(\frac{\alpha_{c,\rm eq}}{\alpha_{\rm s}}\right)^2, \qquad E_b = M_0 - M. \tag{43}$$

In this sequence, $\alpha_{c,\rm eq}$ is the equilibrium central lapse, $z_c = \alpha_{c,\rm eq}^{-1} - 1$ is the central redshift, M_0 is total rest mass, and M is ADM mass [1, 3]. From diffuse, nearly Newtonian clusters ($y_c \to 1$, low z_c) to compact, relativistic cores (smaller y_c, high z_c), E_b/M_0 rises to one maximum along the sequence. At still smaller y_c, E_b decreases and turns negative ($M > M_0$). Like the turning point of fluid stars [57, 58], that binding-energy maximum separates stable clusters at larger y_c from unstable clusters at smaller y_c [11, 12, 21]. Perturbed clusters on the high-y_c branch hold together, whereas clusters past the maximum collapse or expand. Each unit-mass cluster draws up to 10^4 equal-mass macroparticles with zero net momentum (Sec. III F). Static meshes evolve the two loosest models and adaptive meshes evolve the kicked compact ones, with all meshes and particle numbers in Table I.

Multi-pass grid filtering suppresses discrete source fluctuations at $N = 10^4$, yet uncompensated smoothing biases self-gravitating equilibria outward. An unperturbed cluster on the stable branch ($y_c = 0.819$) was expected to stay near equilibrium with minor oscillations, as at $N \geq 10^6$ in Ref. [21]. With four plain binomial passes, the cluster instead expanded by a visible fraction of its radius before rebounding (Fig. 11, left panel, solid curves). During that excursion, central density dropped while α_c rose well above $\alpha_{c,\rm eq}$ (Table V). A first reading, that discrete sampling noise drove this excursion, was ruled out when three independent 10^4-particle samples traced the identical bounce up to a tiny ripple (Fig. 11, left panel). Sweeps across zero, two, and four plain passes on one sample showed that the initial lapse rise grows systematically with pass count $n_{\rm f}$ (Fig. 11, middle panel). Each plain binomial pass widens every cloud's second moment, which acts as a gravitational softening over the filter scale [2] (Sec. III C). Plain passes also lower the proper-volume integral of E slightly across nonuniform $\sqrt{\gamma}$. Because particle velocities match the unsoftened equilibrium, that shallower smoothed potential lets the cluster expand and recontract. The compensation pass of Sec. III C [6] cancels that second-moment softening and preserves the proper-volume energy integral. It removes the bounce across two and four passes and leaves $\alpha_c(t)$ insensitive to $n_{\rm f}$ (Fig. 11, dashed curves). Unfiltered evolution ($n_{\rm f} = 0$) also avoids the bounce, yet leaves constraint norms much higher and causes α_c to wander downward later (Fig. 11, middle panel). All cluster runs

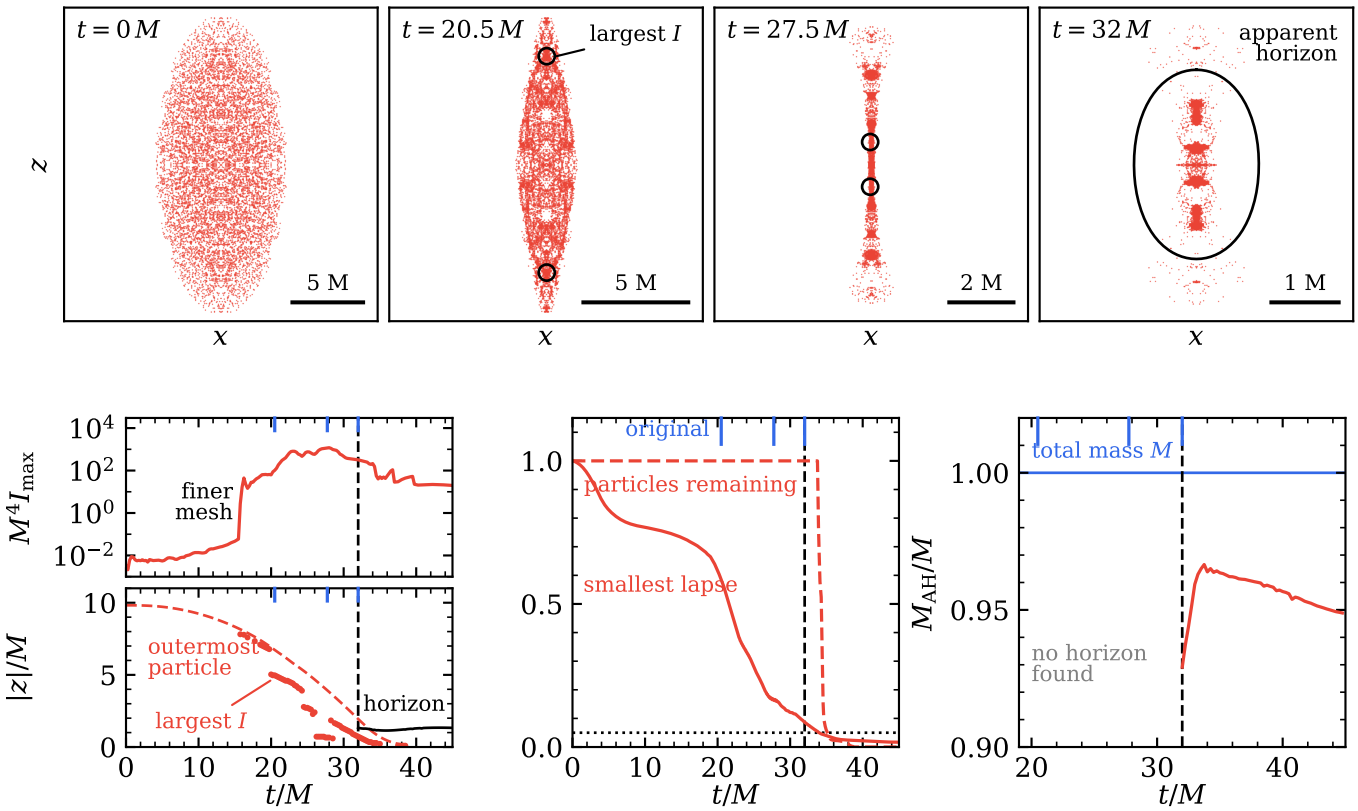


FIG. 10. The cold prolate spheroid ($e = 0.9$, $N = 10^4$) collapses into a segmented spindle whose two axial curvature maxima move inward in steps before an apparent horizon forms. Top: macroparticles in the x–z plane, maxima of the Kretschmann scalar I (circles), and first apparent horizon (black curve). Bottom: peak I and axial distance $|z|$ of that peak (left), smallest lapse and surviving macroparticle fraction (middle), and apparent-horizon mass (right). Dashed: first apparent horizon; dotted: $\alpha_{\rm exc}$; blue: total mass M and the times shown in Ref. [21].

below therefore combine four binomial passes with compensation.

Dynamically stable clusters on the high-y_c side of the binding-energy maximum are expected to remain bound and nearly stationary under small radial perturbations. As in Ref. [21], we apply a one-percent inward velocity kick to the stable models $y_c = 0.900$ and $y_c = 0.819$ at $N = 10^4$ (Fig. 12, first two columns). Over several crossing times, both clusters retain every macroparticle, stay centered near the origin, and hold successive rest-mass radii within a narrow band around equilibrium, as in Ref. [21]. Consistent with its shallower potential, the looser model $y_c = 0.900$ exhibits much smaller variations in α_c and mean Lorentz factor than $y_c = 0.819$ (Table V).

Closer inspection of the stable model $y_c = 0.819$ reveals a coherent radial oscillation of its core that grows before saturating at late times. With filter softening removed, an unperturbed equilibrium was expected to hold a steady central lapse $\alpha_c = \alpha_{c,\rm eq}$. Instead, α_c oscillates at the radial period of a particle in the core while outer shells barely move (Fig. 13, top panel; Table V). Because a nearly uniform core creates an almost harmonic potential with a shared radial period, collective core motion resists phase mixing [2]. Over the first several periods the swing grows steadily, so a long continuation tested unbounded instability against saturation. In that extended run, growth halts after several cycles and the swing stays bounded at a small fraction of $\alpha_{c,\rm eq}$ with all 10^4 particles retained (Fig. 13, top panel). A first reading, that intermediate-wavenumber sampling noise passed by the compensated filter excited this mode, was ruled out by three stratified samples (Fig. 13, top panel) and a Monte Carlo sample. All samples share the same period, phase, and growth. A run at $N = 10^3$ also gives matching early swings instead of scaling as $N^{-1/2}$ (Fig. 13, bottom left).

Further controls on the first sample tested whether time-stepping lag, constraint handling, cloud width, or cell spacing drives this core oscillation (Fig. 13, bottom middle and right; Table I). Frozen sources across Δt could lag particle motion, yet halving the step to $\Delta t/2$ leaves $\alpha_c(t)$ unchanged. Feedback from constraint damping or advected constraint residuals was tested across raised and zero κ_1 and with parabolic relaxation of the Hamiltonian constraint (Sec. III D). Although those settings alter $\|\mathcal{H}\|$ substantially, all three reproduce $\alpha_c(t)$ to a tiny fraction of the swing (Fig. 13, bottom right; Table V). An extended compensation weight c canceling cloud width trims the slow drift slightly (Fig. 13, bottom middle),

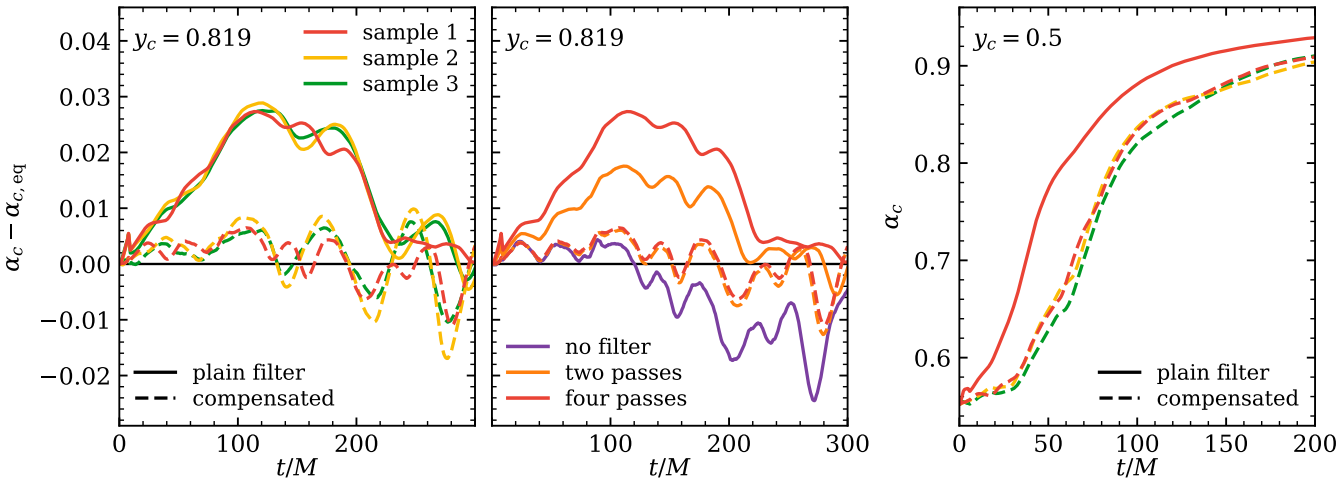


FIG. 11. Source filtering and radial motion of monoenergetic clusters with $N = 10^4$ macroparticles. Left: central-lapse offset $\alpha_c - \alpha_{c,\mathrm{eq}}$ from equilibrium (black line) for the stable model $y_c = 0.819$ across three samples (colors), comparing four plain binomial passes (solid) with compensated filtering (dashed). Middle: one sample with zero, two, and four filter passes (colors), plain (solid) and compensated (dashed). Right: central lapse α_c of the unperturbed unstable model $y_c = 0.5$, encoded as on the left.

and halving the cell size over the cluster likewise leaves the oscillation unchanged. On the full-range lapse axis of the sequence figure in Ref. [21], a swing of this size in $y_c = 0.819$ sits at the resolution limit of the curve. By contrast, the figure for the perturbed near-critical cluster in Ref. [21] at $N \geq 10^6$ displays the same growth, saturation, and decay at the core period. What sets the small common initial offset from which the unperturbed oscillation starts here is not determined.

Smaller counts N at fixed mass M amplify force fluctuations as $N^{-1/2}$ and hasten collisional heating. Stratified and Monte Carlo runs of $y_c = 0.819$ across $N = 10^4$, 10^3, and 10^2 locate where graininess overtakes collective dynamics (Table I). At $N = 10^3$, the cluster holds together and follows the $N = 10^4$ oscillation across several periods (Fig. 13, bottom left). Later cycles at $N = 10^3$ develop deeper lapse minima and a slow downward drift. At $N = 10^2$, discrete fluctuations overtake the cluster within one period. Net momentum wanders, mean kinetic energy more than doubles, and α_c plunges steadily as equilibrium breaks down (Table V). Both the $N = 10^4$ and 10^3 clusters hold across the comparison run, whereas the $N = 10^2$ cluster loses equilibrium.

Past the binding-energy maximum the clusters are unstable [11, 12]. Reference [21] finds collapse under an inward kick for the bound models ($E_b > 0$) and dispersal for the model with $E_b < 0$. With the one-percent inward kick on adaptive meshes, both bound unstable models $y_c = 0.605$ and $y_c = 0.500$ collapse to black holes (Fig. 12, third and fourth columns). Once infall accelerates, mass shells contract together, α_c plunges toward zero, and an apparent horizon forms. As outer shells accrete, M_{AH} rises rapidly to M while lapse excision removes almost every macroparticle (Fig. 12, bottom panels; Table V). Black hole formation and full mass capture agree with Ref. [21], alongside two documented differences. First, apparent horizons form later here for both $y_c = 0.605$ and $y_c = 0.500$ than at $N \geq 10^6$ in Ref. [21]. Reference [21] notes that these collapse times depend on N and sampling noise. In $y_c = 0.605$, a growing core oscillation also precedes collapse (Fig. 12, third column), whereas the published lapse falls smoothly. The cause of these later collapse times is not determined. Second, after horizon formation α_c settles closer to zero here than in Ref. [21]. This difference concerns the coordinates at the center, and its origin was not looked into. For the unbound cluster $y_c = 0.283$ ($E_b < 0$), α_c dips briefly after the kick before rebounding into free dispersal (Fig. 12, rightmost column). Its central lapse approaches unity and mass shells expand at constant asymptotic speeds with every macroparticle kept. This free dispersal agrees with Ref. [21], whereas early spherical runs [12] reported black hole formation.

Without a kick, an unstable cluster past the binding-energy maximum can expand dynamically toward the stable branch of equal mass, a nonlinear transition termed migration [21, 39]. Because the same model $y_c = 0.5$ expands when unperturbed in Ref. [21] whereas kicked $y_c = 0.500$ collapses, we first asked whether plain-filter softening alone selected expansion over collapse. Paired runs of unperturbed $y_c = 0.5$ with plain and compensated four-pass filtering tested that hypothesis directly (Fig. 11, right panel). Plain filtering advances the expansion noticeably, yet three compensated samples still expand outward on a common schedule (Fig. 11, right panel). Plain filtering therefore hastens an already outward transition. In the full run on a static mesh with compensation, the unperturbed cluster $y_c = 0.5$ expands

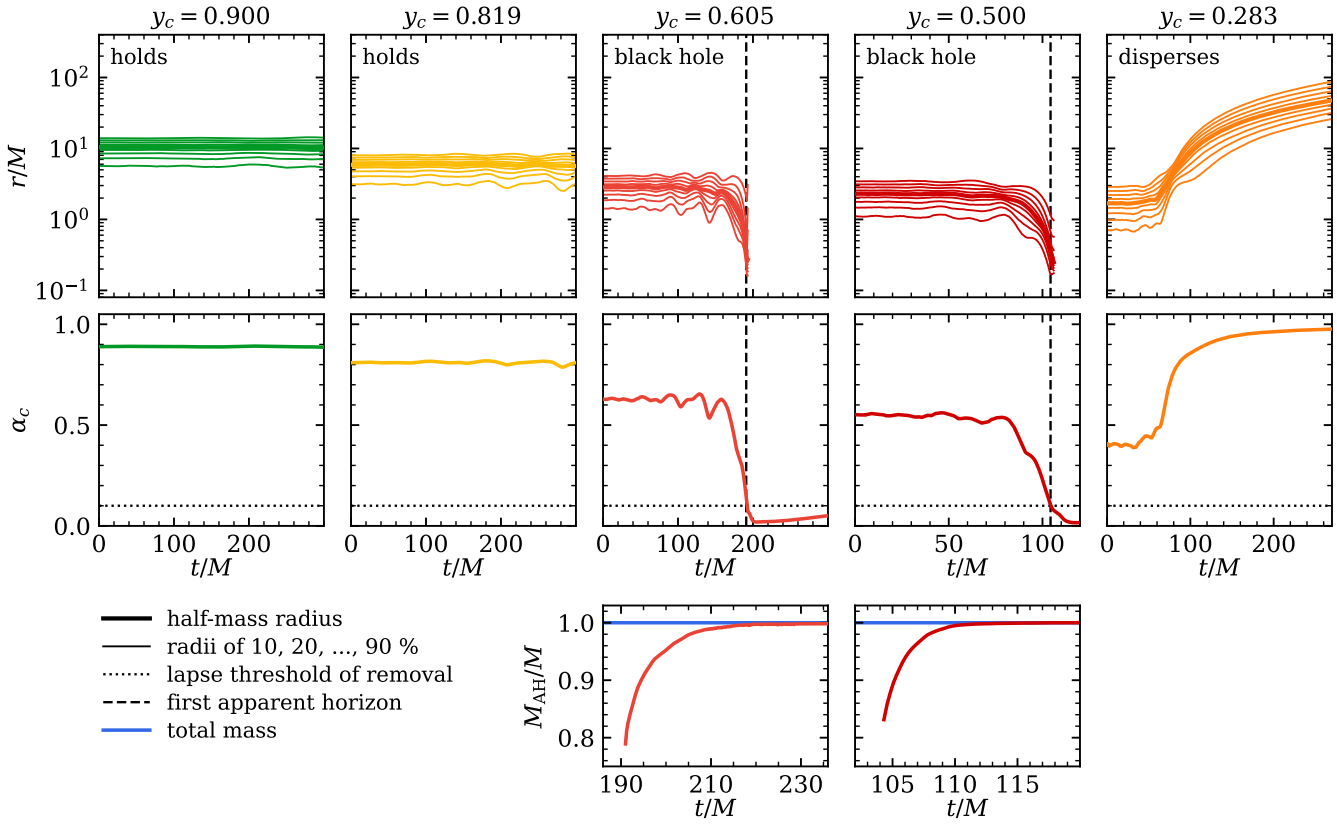


FIG. 12. Five monoenergetic clusters ($N = 10^4$) with an inward velocity kick, from least compact (left) to unbound (right). Models $y_c = 0.900$ and 0.819 hold, 0.605 and 0.500 form black holes, and 0.283 disperses. Top: coordinate radii enclosing tenths of remaining rest mass (half-mass thicker) while at least half remains. Middle: central lapse α_c with excision threshold $\alpha_{\rm exc}$ (dotted) and first apparent horizon (dashed, also top). Bottom: apparent-horizon mass $M_{\rm AH}$ and total mass M (blue).

strongly and then recontracts deeply (Fig. 14; Table I). After that rebound, the cluster settles into a damped oscillation about a more extended state to the end of the run. The sequence of states and the first two maxima and minima of α_c agree with the migration curve of Ref. [21] (Table V). After the first maximum, every extremum of α_c arrives earlier here than in Ref. [21], and the later swings are smaller. The cause of these two differences is not determined. Over the run, a few macroparticles leave the domain and none is removed by lapse excision.

Near the binding-energy maximum, Ref. [21] applies a ten-percent density perturbation to $y_c = 0.756$ and observes bounded oscillations with every macroparticle retained. Its text states the amplitude of the perturbation and leaves its sign and the initial-metric construction open. Here homologous radial scaling changes the density by ten percent of either sign while ψ solves $\mathcal{H} = 0$ for the perturbed matter (Appendix B). Under ten-percent compression, the cluster contracts and collapses to a black hole with every macroparticle removed by lapse excision (Fig. 15; Tables I and V). The final apparent-horizon mass equals the ADM mass of the compressed initial data. A control run with the compressed particles on the unperturbed equilibrium metric contracts in the same way. Under ten-percent expansion, the cluster oscillates and keeps every macroparticle (Fig. 15). Its oscillation runs about a higher central lapse and with a slower rhythm than the published curve of Ref. [21]. The unperturbed model $y_c = 0.756$ oscillates about its equilibrium at the core period with a swing that grows and then stops growing (Fig. 15). Its mean lapse falls slowly below the equilibrium value over the run, and the published curve falls in the same way. That evolution matches the behavior of the stable model in Fig. 13 and resembles the published curve of the perturbed model in Ref. [21] most closely. As an interpretation, the oscillation in the published figure may be this intrinsic core oscillation, while which perturbation Ref. [21] applied cannot be decided from its text.

VI. HOW THE CODE WAS ASSEMBLED

The published text of the numerical-relativity particle-in-cell (NRPIC) method [21] provided the sole specification for the coupled Einstein-Vlasov solver built here

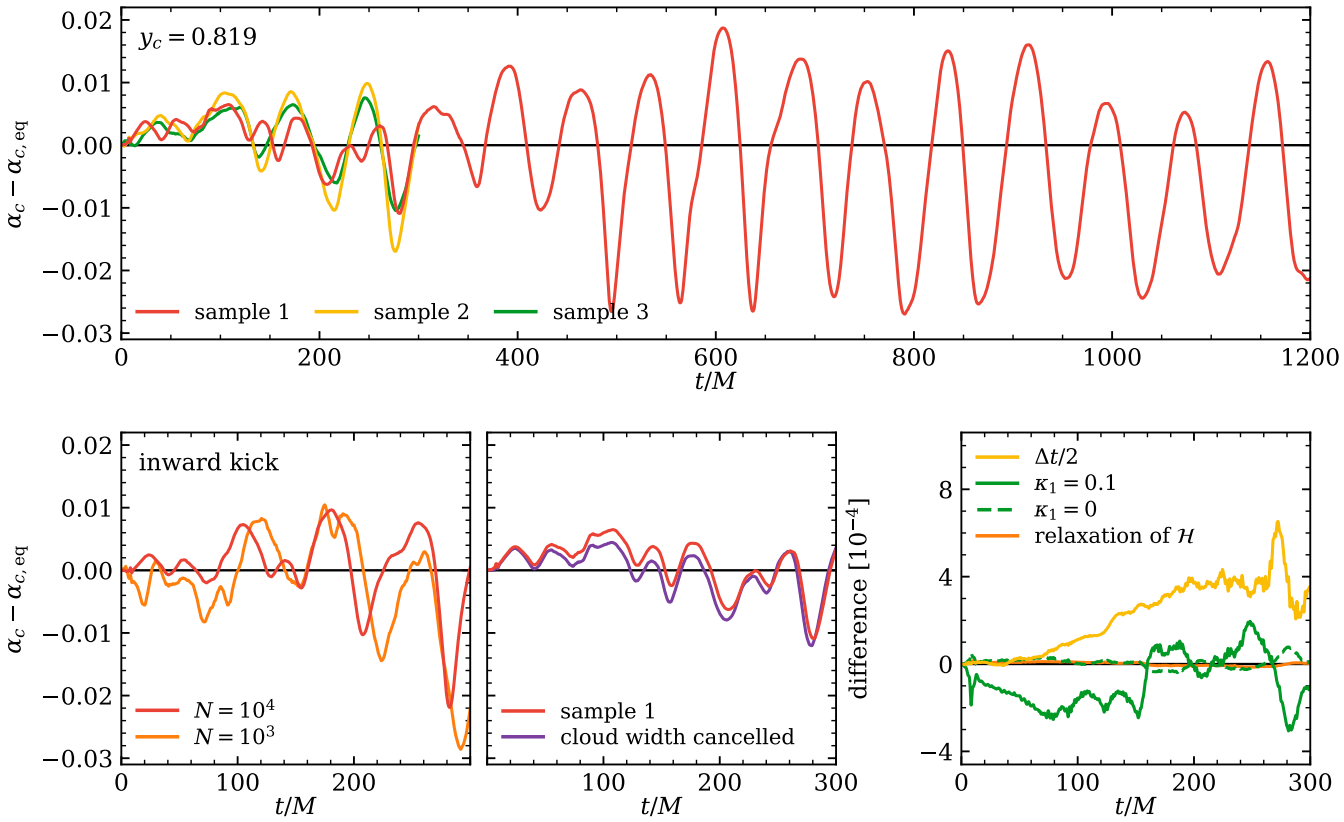


FIG. 13. Core oscillation of the stable cluster $y_c = 0.819$ in central-lapse offset $\alpha_c - \alpha_{c,\mathrm{eq}}$ from equilibrium (black lines). Top: three unperturbed samples at $N = 10^4$ (colors), with sample 1 continued to saturation. Bottom left: inward-kicked runs at $N = 10^4$ and 10^3. Bottom middle: sample 1 against compensation that also cancels cloud width. Bottom right: differences from sample 1 for $\Delta t/2$, raised and zero (dashed) constraint damping κ_1, and relaxation of the Hamiltonian constraint.

without access to the original authors' code. This independent reconstruction tests whether the written equations and algorithmic rules fix the coupled method completely. The implementation assembles the coupled solver inside the host code AthenaK [35, 36] from existing capabilities across neighboring fields. AthenaK supplies the block-structured octree mesh with adaptive refinement, ghost-zone exchange, prolongation, and restriction. It also provides the Z4c spacetime solver in moving-puncture gauge, the particle container and drift update, the magnetohydrodynamics matter-source interface, and the pseudospectral apparent-horizon finder [35, 36]. Portable parallel execution of mesh stencils and particle loops comes from the Kokkos performance-portability library [37].

Plasma particle-in-cell methods contribute implicit midpoint geodesic integration [27], the velocity-space Boris rotation [6, 26], cloud-in-cell deposition with binomial filtering and compensation [6, 22], and conservative single-level deposition across refinement boundaries [30]. Numerical relativity and mesh methods supply the tetrad form of geodesic motion [28, 29], lapse excision inside black holes [49], parabolic relaxation of the Hamiltonian constraint [45, 46], and the Löhner refinement indicator [48]. Exact analytical solutions for Kerr geodesics, Oppenheimer–Snyder dust collapse [38], monoenergetic cluster equilibria [1], and homogeneous spheroid potentials complete the external inputs. This work combines those inputs into the two geodesic pushers in the symmetric and Cholesky tetrad gauges, cubic Hermite metric reconstruction, and level-capped deposition with compensated filtering. Additional components built here provide particle-coupled adaptive mesh refinement, lapse excision with interpolant-derivative apparent-horizon diagnostics, constraint-control terms, initial-data and validation suites, and the staggered coupled cycle.

Figure 16 shows which capabilities each component combines. Horizontal rows list the existing capabilities grouped along the left margin by their origin. Vertical red columns represent the components built in this work as hyperedges of a hypergraph. Blue dots along each column mark the inherited capabilities joined by that hyperedge. Red rows at the bottom denote the geodesic-pusher and deposition components built here. Red dots on those bottom rows mark their reuse inside the adaptive-mesh and coupled-cycle hyperedges.

Large-language-model agents carried out the reconstruction under human direction across dedicated solver-

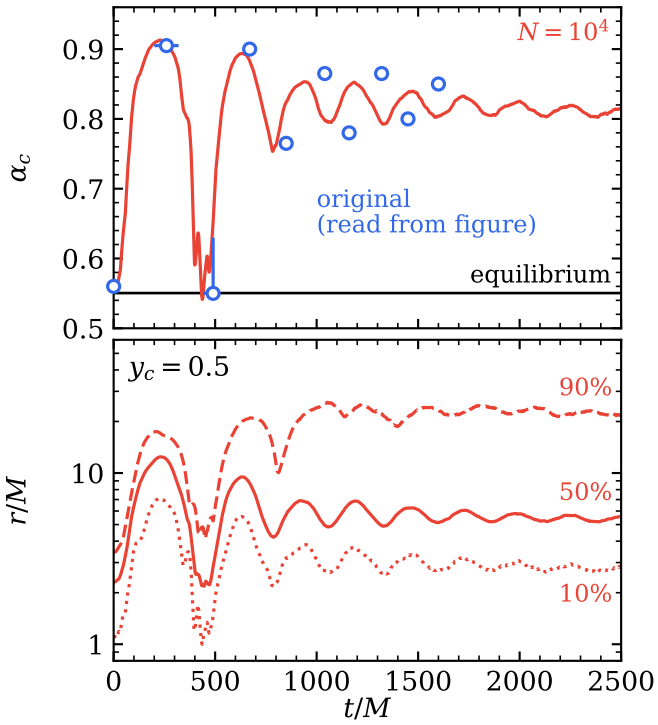


FIG. 14. The unperturbed cluster $y_c = 0.5$ expands, recontracts deeply, and oscillates with decreasing swing about a more extended state. Top: central lapse α_c, equilibrium (black), and extrema from Ref. [21] (circles; bars: read ranges). Bottom: radii enclosing the labeled rest-mass fractions.

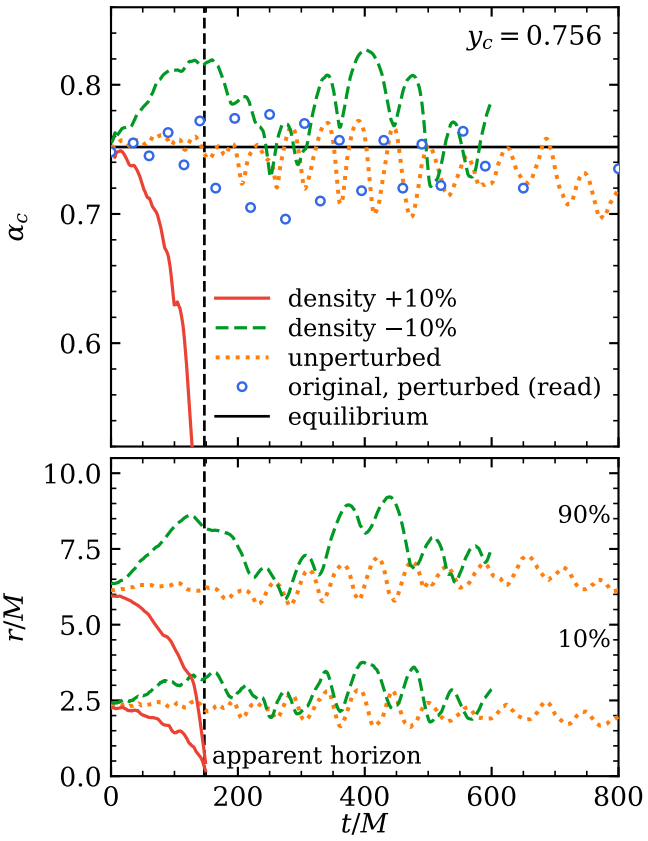


FIG. 15. The near-critical cluster $y_c = 0.756$ collapses to a black hole when compressed, oscillates when expanded, and shows the core oscillation when unperturbed. Top: central lapse α_c, first apparent horizon (dashed), and extrema from Ref. [21] (circles). Bottom: radii enclosing the labeled rest-mass fractions.

development, verification, and manuscript roles. A single language model in the solver-development role produced the component design, the symbolic derivations, and the solver source. Separate language models in the verification role handled builds, simulation runs, output analysis, and exact reference solutions that were written without access to the solver source. An independent language model in the manuscript role wrote the text of this paper from the method specifications, figures, and test outcomes.

Candidate explanations for unexpected simulation results were tested against predictions recorded in advance. Each diagnostic run meant to explain an anomaly was preceded by a written statement of the outcome expected under every proposed mechanism. The initial explanation for an apparent-horizon mass below M after spherical collapse held that the pseudospectral flow finder stopped its iteration too early. Calibration runs on an evolved vacuum black hole of known mass M tested that explanation with a tighter iteration stopping rule. The tighter stopping rule left the reported apparent-horizon mass unchanged. Independent evaluations of the slice Misner–Sharp mass M_{MS} confirmed that the evolved black hole spacetime carried the mass M to the accuracy shown in Fig. 9. Direct inspection of the host code's finder showed that it interpolates a cell-centered metric-derivative array whose ghost zones retain their allocated zeros across mesh-block faces. Differentiation of the ghost-filled metric interpolant on the same saved slices brought the apparent-horizon mass into agreement with M_{MS} across the plotted resolutions. An independent agent rebuilt the appendix tables of setups and outcomes directly from the saved simulation outputs and exposed unsupported log entries that were then corrected. Recomputation of deposition from stored particles confirmed that a central-density excess under coarser particle clouds, first logged as agreement with the cycloid, arose from the regular particle lattice beating against the cloud lattice. Several other initial explanations failed analogous predictive tests across the scenarios of Sec. V. Every effect without an isolating run has its cause stated there as undetermined.

All materials needed to inspect and extend the simulations are public alongside the solver source code. The public release includes the simulation inputs, the analysis scripts, and the chronological record of every prediction

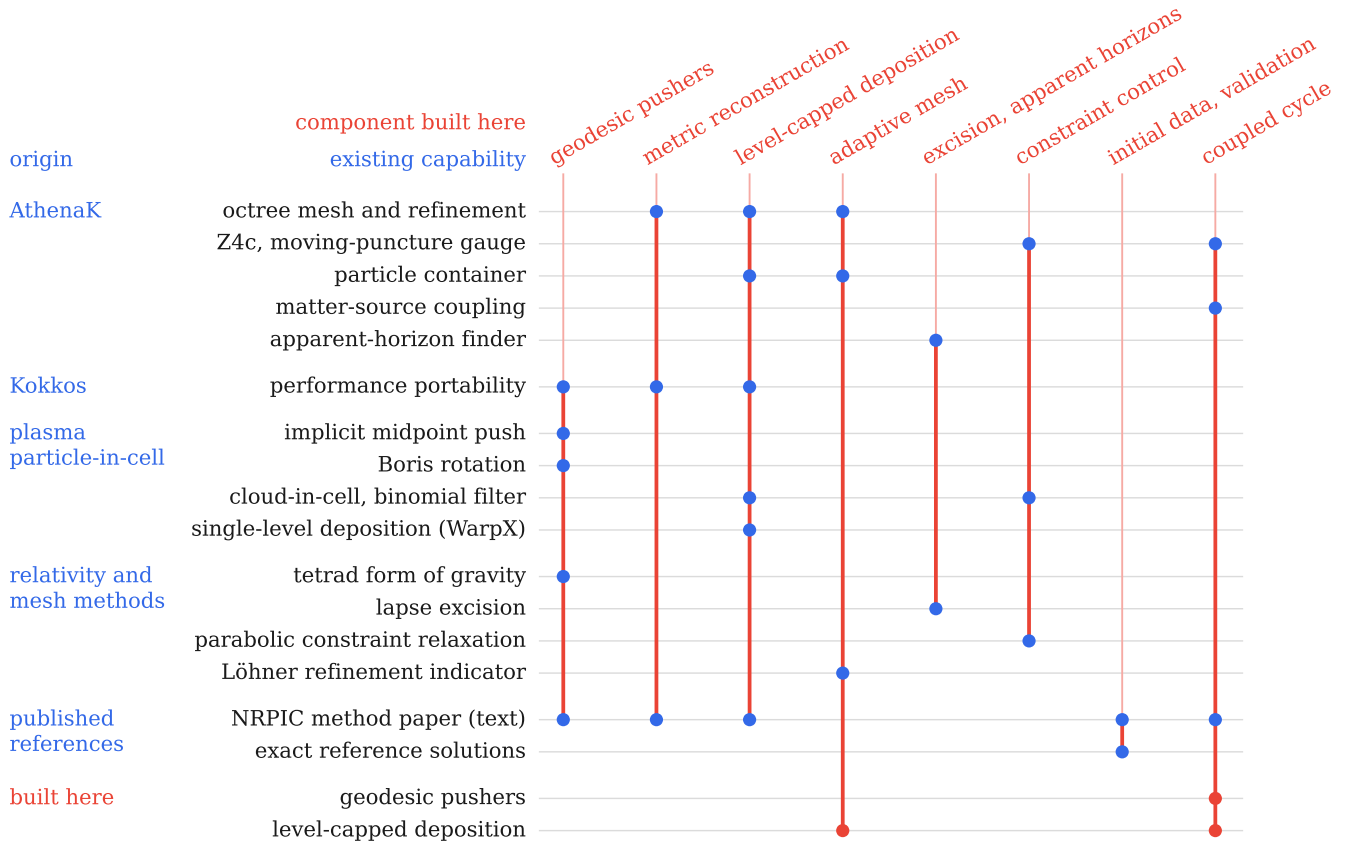


FIG. 16. The hypergraph shows how the code was assembled. Each row is a capability taken from an existing code or published method, grouped by origin. Each red column is a component built here, a hyperedge joining the capabilities it combines (dots). The red rows at the bottom are built components reused inside the coupled cycle and the adaptive mesh.

and outcome. Readers can rerun the validation benchmarks, modify individual stages of the particle-mesh cycle, or audit each diagnostic test from those materials. The data-availability statement at the end of the paper gives the archive location.

DATA AND CODE AVAILABILITY

The solver source (a particle module inside the public code AthenaK [35, 36]), the inputs of every run, and the initial-data generators are public at `https://github.com/FoAKTEE/nrpic`. The same repository provides the analysis and figure scripts alongside the data behind every figure. It also holds the ledger of every number in the tables and the chronological record of predictions and outcomes.

ACKNOWLEDGMENTS

This work rebuilds the method of Ref. [21] from its published text, and we thank its authors for the level of detail of that text. We acknowledge the developers of AthenaK [35, 36] and Kokkos [37] for the public host code and performance-portability library used here. Large-language-model agents under human direction wrote the solver source, ran and analyzed the simulations, made the figures, and wrote the text. One language-model family served each of the dedicated solver-development, verification, and manuscript roles described in Sec. VI. The human authors directed the work and decided what was tested. They take responsibility for the content of this paper.

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Appendix A: Scenario parameters and measured values

Table I lists the setup of every family of runs. Tables II and V list the measured values to which the text refers, with the values of Ref. [21] where it gives them.

TABLE I. Setups, one row per family of runs: model, macroparticle number N and sampling, domain half-width L, mesh and cell spacing Δx in the matter at the start and the finest reached (vacuum runs: at the puncture), first time step Δt, filter passes n_f (+c: compensation pass), deposition, excision threshold α_exc, constraint damping κ_1, final time. Meshes other than uniform have 64^3 root cells in MeshBlocks of 16^3 cells. Parentheses: single runs of a family.

Scenario and model	N, sampling	L/M	Mesh, $\Delta x/M$	$\Delta t/M$	n_f	Deposition	α_exc	κ_1	t_end/M
Kerr geodesics: $a = 0.9M$, $a \to 0$; bound, $e = 0.99$, plunging	1	10–12	uniform, 1/8–1/2; lattice 0.025	0.00125–0.08	–	–	–	–	77.5–4550
Test-particle clusters: $a = 0.9M$, $a \to 0$; cloud at $r_0 = 10M$	10^2–10^4, Gaussian	64	static, 1/4→1/16	0.05	2	finest level	0.62	–	500–1000
Vacuum black hole: puncture of mass M; 6 tracers	6, tracers	128	static, 1/8 (1/4, 1/16)	1/32 (1/16, 1/64)	–	none	0.05	0.02	12–250
Head-on collision: punctures $M/2$ at $x = \pm 5M$; 6 tracers	6, tracers	128	adaptive, 1/16	1/64	–	none	0.05	0.02	160
Dust collapse: $R_0 = 5M$, $h = 0.30M$; masses corrected, published	9328, lattice	155	adaptive, 0.30→0.038	0.076	1	finest level	0.1	0.02	60
Dust collapse, level cap: as above, corrected masses	9328, lattice	155	adaptive, 0.30→0.038	0.076	1	level-capped (≤ 1, ≤ 2, any)	0.1	0.02	60
Dust collapse, random sample: $R_0 = 5M$; equal masses[d]	9328, Monte Carlo	155	adaptive, 0.30→0.038 (0.30→0.076)	0.076	1, 4+c	level-capped (finest level)	0.1	0.02	41–60[c]
Prolate collapse: spheroid $e = 0.9$, $b = 10M$, at rest	10^4, Monte Carlo	128	adaptive, 1/4→1/16	1/16	4+c	finest level	0.05	0.02	45
Stable cluster, filter: $y_c = 0.819$, unperturbed; samples 1–3	10^4, stratified	1024	static, 1/2	1/8	0, 2, 2+c, 4, 4+c	finest level	0.1	0.02	300
Stable cluster, oscillation: $y_c = 0.819$, unperturbed; sample 1	10^4, stratified	1024	static, 1/2 (1/4–1/2)	1/8 (1/16)	4+c[b]	finest level	0.1	0.02 (0.1, 0)[a]	300–1200
Particle number: $y_c = 0.819$; kick 0.01 or none	10^2–10^4, Monte Carlo, stratified	1024	static, 1/2	1/8	4+c	finest level	0.1	0.02	300–500
Sequence, stable: $y_c = 0.900, 0.819$; kick 0.01	10^4, stratified	1024	static, 1/2	1/8	4+c	finest level	0.1	0.02	300
Sequence, unstable: $y_c = 0.605, 0.500, 0.283$; kick 0.01	10^4, stratified	512	adaptive, 1/4→1/16	1/16	4+c	finest level	0.1	0.02	120–300
Unperturbed unstable: $y_c = 0.500$, unperturbed; samples 1–3	10^4, stratified	512	static, 1/4	1/16	4, 4+c	finest level	0.1	0.02	200–2500
Near-critical: $y_c = 0.756$; density $\times(1 \pm 0.1)$ or unperturbed	10^4, stratified	512 (1024)	static, 1/4 (adaptive, 1/4→1/16; adaptive, 1/4)	1/16	4+c	finest level	0.1	0.02	160–818

[a] One run adds parabolic relaxation of $\mathcal{H}$ with $c_H = 0.05\,M$.
[b] One run uses $c = n_\mathrm{f}/4 + 1/12$, which also cancels the cloud width.
[c] Two capped runs with the full refinement depth stopped at the time limit of their jobs, at $t = 43M$ and $t = 41M$.
[d] With the published rule at the full refinement depth the MeshBlock count passed the limit before a horizon formed: $t = 8.8M$, 1884 asked for, limit 1800 ($n_\mathrm{f} = 1$); $t = 10.2M$, 2024 asked for, limit 2000 ($n_\mathrm{f} = 4 + c$).

Appendix B: Initial data and sampling

Coupled Einstein-Vlasov evolutions require constraint-satisfying initial slices alongside discrete particle ensembles that sample the continuum phase-space distribution. Each self-gravitating run starts on a conformally flat, time-symmetric slice where superposed flat-space Poisson potentials u_c for conformal sources $\tilde{\rho}_c$ solve $\mathcal{H} = 0$ via Eq. (36). Around each component center $\boldsymbol{x}_c$ at isotropic distance $\bar{r}_c = |\boldsymbol{x} - \boldsymbol{x}_c|$, four component families enter Eq. (36). Tabulated spherical profiles interpolate

TABLE II. Measured quantities to which the text refers for test particles and the apparent-horizon check, with the values of Ref. [21] where it gives them (from its text, or read from its figures). Masses in units of the total mass M, times in units of M; listed values follow the order named in the first column.

Quantity	This work	Ref. [21]
Fixed black hole		
Passage $e = 0.99$, largest relative errors of $\mathcal{E}$, L_z, Q ($\Delta t = 0.01M$, to $230M$):		
implicit midpoint, analytic metric	2.78×10^{-8}, 0, 1.25×10^{-7}	2.8×10^{-8}, round-off, 1.3×10^{-7}
spacetime Boris, symmetric gauge, analytic	9.05×10^{-8}, 1.14×10^{-8}, 7.83×10^{-7}	9.0×10^{-8}, 1.1×10^{-8}, 7.8×10^{-7}
implicit midpoint, Hermite lattice	2.16×10^{-6}, 5.30×10^{-6}, 5.06×10^{-5}	2.2×10^{-6}, 5.3×10^{-6}, 5.1×10^{-5}
spacetime Boris, Cholesky gauge, Hermite lattice	2.05×10^{-6}, 3.79×10^{-6}, 5.29×10^{-5}	2.0×10^{-6}, 3.8×10^{-6}, 5.3×10^{-5}
Passage, order of convergence in Δt (three pushers, analytic metric)	1.995–2.001	second order
Inclined Kerr orbit on the mesh, order in h of position and $\mathcal{E}$ errors	2.89, 3.29	–
Plunging clusters ($N = 10^4$): removed by excision, against exact geodesics; $a \to 0$ and $a = 0.9M$	3726 vs 3727; 3345 vs 3339	–
Bound clusters, exponent of the filtered-E differences against N; $a \to 0$ and $a = 0.9M$	-0.48, -0.48	–
Evolving black holes		
Puncture: largest relative change of L_z of the four circular tracers	1.1×10^{-4}–3.9×10^{-4}	–
Puncture: radial infall and plunge removed at $\alpha_{\rm exc}$, t/M	54, 126	–
Puncture ($\Delta x = M/8$), slice Misner–Sharp horizon mass at $t = 50M$	1.0016	–
Head-on: first common apparent horizon t/M; its mass at the end	49.25; 1.0026	–
Head-on: largest relative change of L_x (five orbiting tracers); $\lvert L_x\rvert$ of the tracer from rest	9.8×10^{-4}; 4×10^{-14}	–
Apparent-horizon check		
Finder as distributed, $t = 50M$, at $\Delta x = M/4$, $M/8$, $M/16$	0.9353, 0.9771, 0.9406	–
Slice Misner–Sharp horizon mass, same slices	0.9656, 1.0016, 1.0000	–
Finder as distributed ($M/8$), stopping tolerance 10^{-5} and 10^{-8}, $t \simeq 50M$	0.9771, 0.9771	–
Corrected finder in a live run ($M/8$), last search; slice mass at $t = 12M$	1.0003; 1.0006	–

$u_c(\bar r_c) = \psi_c(\bar r_c) - 1$ on a uniform grid with four-point Lagrange polynomials and follow Eq. (37) outside with the component ADM mass. Vacuum black hole punctures evaluate Eq. (37) directly, while spherical Gaussian components of conformal mass m_c and width σ_c obey

$$\tilde\rho_c(\bar r_c) = \frac{m_c}{(2\pi\sigma_c^2)^{3/2}} \exp\!\left(-\frac{\bar r_c^2}{2\sigma_c^2}\right),$$
$$u_c(\bar r_c) = \frac{m_c}{2\bar r_c} \operatorname{erf}\!\left(\frac{\bar r_c}{\sqrt{2}\,\sigma_c}\right). \tag{B1}$$

A homogeneous prolate spheroid of ADM mass M, polar semi-axis b along z, eccentricity e, equatorial semi-axis $a_{\rm eq} = b\sqrt{1-e^2}$, and focal half-distance $d_{\rm f} = eb$ has closed-form fields [54, 56]

$$\tilde\rho_c(\boldsymbol{x}) = \frac{3M}{4\pi a_{\rm eq}^2 b}, \qquad \frac{x^2+y^2}{a_{\rm eq}^2} + \frac{z^2}{b^2} \le 1,$$
$$u_c(\boldsymbol{x}) = \frac{3M}{8}\Big[I_0(\lambda) - (x^2+y^2)J_{\rm eq}(\lambda) - z^2 J_z(\lambda)\Big],$$
$$I_0(\lambda) = \frac{2}{d_{\rm f}} \operatorname{arctanh}\!\left(\frac{d_{\rm f}}{b_\lambda}\right),$$
$$J_{\rm eq}(\lambda) = \frac{b_\lambda}{d_{\rm f}^2(a_{\rm eq}^2+\lambda)} - \frac{I_0(\lambda)}{2d_{\rm f}^2},$$
$$J_z(\lambda) = \frac{I_0(\lambda)}{d_{\rm f}^2} - \frac{2}{d_{\rm f}^2 b_\lambda},$$
$$1 = \frac{x^2+y^2}{a_{\rm eq}^2+\lambda} + \frac{z^2}{b^2+\lambda} \qquad (\lambda > 0). \tag{B2}$$

The auxiliary length $b_\lambda = \sqrt{b^2+\lambda}$ sets $\lambda = 0$ inside the spheroid, while outside the boundary $\tilde\rho_c$ vanishes and $\lambda > 0$ solves the last relation. Every initial slice sets vanishing

TABLE III. As Table II, for dust collapse. $\|\mathcal{H}\|$ is the square root of the integral of $\mathcal{H}^2$ over the proper volume of the grid, cells with $\chi < 1/16$ excluded; the same measure is used in every table.

Quantity	This work	Ref. [21]
Dust collapse on a lattice		
Central proper time τ_c/M at which α_c exceeds the lapse bound by 10^{-2}	3.53	$\simeq 3.5$
Rest mass $\sum_p m_p/M$: published lattice rule; surface-corrected (continuum M_0)	1.1687; 1.1551	$\simeq 1.155$ ($N \simeq 10^6$)
First apparent horizon t/M (finest-level clouds, corrected masses)	16.39	–
$M_{\rm AH}/M$ at formation; at $t = 60M$ (finest-level clouds, corrected masses)	1.0004; 0.9987	$\lvert M_{\rm AH}/M - 1\rvert \lesssim 10^{-3}$ at $60M$
$M_{\rm AH}/M$ at formation; at $t = 60M$ (published lattice masses)	1.0114; 1.0099	–
$M_{\rm AH}/M$ when the last macroparticle is removed → at $t = 60M$: finest level; level cap with any, ≤ 2, ≤ 1 levels of coarsening	1.0004→0.9987; 1.0006→1.0065; 1.0006→1.0064; 1.0012→0.9904	–
Largest $\|\mathcal{H}\|$, same four runs	1.9, 4.7, 4.5, 2.8	–
Largest $\|\mathcal{H}\|$ within $1M$ after the refinement to $\Delta x = h/4$, same four runs	1.59, 0.48, 0.48, 1.14	–
Central reading of E (eight central cells, after the filter) over the cycloid at the same τ_c, in the $3M$ after the refinement to $h/4$: finest level; level cap with any, ≤ 1 levels of coarsening	0.27–0.52; 1.11–1.14; 1.28–1.49	–
Dust collapse from a random sample		
Random sample, order of the runs below: capped clouds at the full depth with $n_{\rm f} = 1$ and 4+c; one level fewer, finest-level and capped clouds (4+c)		
first apparent horizon t/M, $M_{\rm AH}/M$	16.35, 0.9949; 16.31, 0.9939; 16.20, 1.0019; 16.58, 0.9990	–
$M_{\rm AH}/M$ at $t = 20M$, $40M$, and over the last $2M$ (last search t/M); medians over $\pm 1M$, as successive searches scatter by up to 10^{-3}	1.0000, 1.0004, 1.0000 (43); 1.0004, 1.0013, 1.0013 (41); 0.9983, 0.9971, 0.9973 (60); 0.9984, 0.9946, 0.9946 (60)	–
Largest MeshBlock count: the four runs above; finest-level clouds at the full depth with $n_{\rm f} = 1$ and 4+c, count asked for when the run stopped (t/M)	1779, 1681, 400, 400; 1884 (8.8), 2024 (10.2)	–
One level fewer, largest $\|\mathcal{H}\|$ before the first horizon, from it until the last macroparticle is removed, and afterwards: finest-level; capped clouds	1.77, 2.93, 0.91; 2.95, 3.79, 2.23	–

TABLE IV. As Table II, for the collapse of a prolate spheroid; $\|\mathcal{H}\|$ as in Table III.

Quantity	This work	Ref. [21]
Prolate collapse		
Largest Kretschmann scalar: time t/M and $\lvert z\rvert/M$ (on the axis)	27.75; 0.66	–
First apparent horizon t/M; $M_{\rm AH}/M$	32.00; 0.929	32 (from its text)
Largest $M_{\rm AH}/M$ (t/M); $M_{\rm AH}/M$ at the last search (t/M)	0.9666 (33.75); 0.9489 (44.75)	–
First and last removal of macroparticles t/M (macroparticles left)	33.78; 38.80 (0)	–
Misner–Sharp mass on the last slice, at the horizon ($\pm$ its spread over angle) and at large radius (t/M of the slice)	0.920 ± 0.027; 0.999 (45)	–
Largest $\|\mathcal{H}\|$ (t/M); lattice dust collapse (finest level, corrected masses) on the same measure	6.31 (27.5); 1.90	–

shift $\beta^i = 0$, with initial lapse $\alpha = 1$ for cold collapse and $\alpha = \psi^{-2}$ for vacuum punctures. Isolated clusters use their static equilibrium lapse joined outside to $(1-u_c)(1+u_c)^{-1}$, whereas the composite cluster uses $\alpha = 2\psi^{-1} - 1$. Since $\sqrt{\gamma} = \psi^6$, N equal-mass macroparticles sampling proper rest-mass density ρ_0 and continuum rest mass M_0 have coordinate number density

$$\mathcal{N}(\boldsymbol{x}) = \frac{N}{M_0}\,\rho_0(\boldsymbol{x})\psi(\boldsymbol{x})^6, \qquad m_p = \frac{M_0}{N}, \qquad M_0 = \int \rho_0(\boldsymbol{x})\psi(\boldsymbol{x})^6\,\mathrm{d}^3x. \tag{B3}$$

For the cold prolate spheroid ($u_{i,p} = 0$, $\rho_0\psi^6 = \tilde{\rho}_c\psi$), one quarter of the N particles are drawn inside the bound-

TABLE V. As Table II, for the relativistic clusters; $\|\mathcal{H}\|$ as in Table III. Central lapse α_c in the eight central cells; $\alpha_c(0)$ is its value at $t = 0$.

Quantity	This work	Ref. [21]
Stable cluster $y_c = 0.819$: filter		
Rise of α_c above $\alpha_c(0)$ within $130M$, sample 1, $n_f = 0$, 2, 4, 2+c, 4+c	0.0044, 0.0176, 0.0273, 0.0061, 0.0065	≈ 0.01 (read from figure)
Same rise, samples 1–3, four plain passes; with compensation	0.0273, 0.0289, 0.0275; 0.0065, 0.0084, 0.0061	–
Filtered over deposited proper-volume integral of E, minus one: four plain passes; with compensation	-3.0×10^{-3}; -1.0×10^{-4}	–
$\|\mathcal{H}\|$ at $t = 300M$: no filter; four plain passes; with compensation	0.141; 0.011; 0.031	–
Stable cluster $y_c = 0.819$: core oscillation		
Period of α_c (100–1200M); radial period in the core, $\pi[3/(4\pi E_c)]^{1/2}$	75; 66	–
Half the swing of α_c in windows of $100M$: growth phase (100–500M); after $500M$	0.005→0.018; 0.015–0.022	–
Largest $\|\Delta\alpha_c\|$ from sample 1 over $300M$: $\Delta t/2$, $\kappa_1 = 0.1$, $\kappa_1 = 0$, c_H	6.5×10^{-4}, 3.1×10^{-4}, 7.9×10^{-5}, 2.1×10^{-5}	–
Same, compensation that also cancels the cloud width	3.6×10^{-3}	–
Near-critical $y_c = 0.756$ in Ref. [21]: period of α_c; here the core radial period of that model	45	≈ 50 (read from figure)
Stable cluster $y_c = 0.819$: particle number		
Monte Carlo, $N = 10^3$ against 10^4: largest difference of $\alpha_c - \alpha_c(0)$ to $190M$; lowest $\alpha_c - \alpha_c(0)$ after $190M$ (10^4, 10^3); mean over 400–500M (10^4, 10^3)	0.005; -0.023, -0.035; -0.003, -0.017	–
Monte Carlo $N = 10^2$: α_c at $t = 0$, $200M$, $500M$; mean Lorentz factor at start and end; largest $\|\sum_p m_p u_{i,p}\|/M_0$	0.81, 0.68, 0.58; 1.047, 1.138; 0.039	–
Cluster sequence with inward kick		
α_c range over $300M$, $y_c = 0.900$; 0.819; particles kept	0.886–0.892; 0.787–0.819; all	≈ 0.89; 0.81–0.82 (read from figure)
First apparent horizon t/M: $y_c = 0.605$; 0.500	191.0; 104.3	109; 73
$M_{\rm AH}/M$ at the end; macroparticles left of 10^4	1.0000; 1.0001; 6, 3	$\simeq 1$ (stated)
α_c at the end, after the horizon forms: $y_c = 0.605$; 0.500	0.051; 0.016	≈ 0.20 (read from figure)
$y_c = 0.283$: α_c at the end (t/M); particles kept	0.976 (270); all	≈ 0.98 (300; read from figure)

ary by rejection against $\psi(\boldsymbol{x})\psi(\boldsymbol{0})^{-1}$ and completed by $(x, y, -z)$ and $(-x, -y, \pm z)$. Gaussian components draw positions around $\boldsymbol{x}_c$ by randomized stratification in radius and angles. In multi-component superpositions, each component multiplies its isolated particle masses by $\psi\psi_c^{-1}$ ($\psi_c = 1$ and rest mass $m_c W_c^{-1}$ at bulk Lorentz factor W_c) and lowers orthonormal momenta with ψ^2.

The homogeneous dust ball of mass M, areal radius R_0, and density ρ_0 (Sec. V A) starts from rest ($u_{i,p} = 0$) with isotropic radius $\bar{r}_0$ and interior conformal factor [53]

$$\bar{r}_0 = \frac{R_0}{2}\left(1 - \frac{M}{R_0} + \sqrt{1 - \frac{2M}{R_0}}\right),$$
$$\psi(\bar{r}) = \sqrt{\frac{\left(1 + \sqrt{1 - 2M/R_0}\right)\bar{r}_0 R_0^2}{2\bar{r}_0^3 + M\bar{r}^2}} \qquad (\bar{r} \le \bar{r}_0), \quad \text{(B4)}$$

matched to Eq. (37) outside. The cube $[-\bar{r}_0, \bar{r}_0]^3$ is partitioned into $n_{\rm lat}^3$ cubic cells of spacing $h = 2\bar{r}_0 n_{\rm lat}^{-1}$ aligned with the initial matter-level mesh, with $n_{\rm lat}$ in the parameter table. Macroparticles occupy cell centers with $\bar{r}_p = |\boldsymbol{x}_p| \le \bar{r}_0$ for noise-free interior deposition, and the published rule [21] assigns proper cell masses

$$m_p^{\rm pub} = \rho_0 \psi(\bar{r}_p)^6 h^3. \qquad \text{(B5)}$$

The count of lattice sites inside a sphere differs from its volume in cells h^3 by a boundary fraction of order $n_{\rm lat}^{-1}$, so summed $m_p^{\rm pub}$ exceeds M_0 on small lattices. Our surface correction leaves interior masses unchanged for $\bar{r}_p \le \bar{r}_0 - h$ and rescales the outermost layer $\bar{r}_0 - h < \bar{r}_p \le \bar{r}_0$ of width h by one common factor so all particle masses sum to M_0,

$$m_p = m_p^{\rm pub}\,\frac{M_0 - \sum_{\bar{r}_{p'} \le \bar{r}_0 - h} m_{p'}^{\rm pub}}{\sum_{\bar{r}_{p'} > \bar{r}_0 - h} m_{p'}^{\rm pub}}. \qquad \text{(B6)}$$

Monoenergetic equilibrium clusters [1] populate a single conserved energy per unit rest mass $E_0 = \alpha_s$ in Eq. (43) with isotropic momentum directions. In dimensionless areal radius $\tilde{r}$, squared lapse ratio $y(\tilde{r})$ with $y(0) = y_c$, and enclosed gravitational and rest-mass variables $\tilde{m}$ and $\tilde{m}_0$, the structure equations and isotropic-

TABLE VI. As Table V, for the unperturbed model past the binding-energy maximum and the near-critical model.

Quantity	This work	Ref. [21]
Unperturbed $y_c = 0.5$		
Time t/M at which α_c first reaches 0.8: plain filter; compensated, samples 1–3	59; 88, 86, 91	≈ 120 (read from figure)
Migration: α_c at the start and at its extrema to $1000M$ (t/M)	0.552 (0); 0.913 (228); 0.591 (400); 0.637 (417); 0.541 (436); 0.894 (631); 0.753 (783); 0.853 (940)	0.56 (0); 0.905 (200–320); 0.55–0.63 (440–520); 0.90 (670); 0.765 (830–880) (read from figure)
Migration: extrema after $1000M$ (t/M); α_c at the end; mean over 1500–2500M	0.795 (1072); 0.853 (1184); 0.792 (1329); 0.840 (1472); 0.802 (1582); 0.833 (1726); 0.804 (1852); 0.826 (2001); 0.816 (2500); 0.814	0.865 (1040); 0.78 (1160); 0.865 (1320); 0.80 (1450); 0.85 (1600) (read from figure); late mean 0.834 (from its text)
Migration: macroparticles removed by excision; leaving the domain (at t/M)	0; 4 (1816, 1999, 2405, 2440)	–
Near-critical $y_c = 0.756$		
Compressed by ten percent: $\alpha_c(0)$; first apparent horizon t/M	0.749; 147.2	$\simeq 0.75$; none
Compressed: $M_{\rm AH}/M$ at the end; M of the initial data; macroparticles left	0.9970; 0.9969; 0	all kept
Compressed: largest $\lvert\Delta\alpha_c\rvert$ to $120M$ between the consistent and the unperturbed initial metric	0.009	–
Density lowered by ten percent: α_c range; mean over $600M$; radii enclosing 90 and 10 percent of the rest mass, range $(t = 0)$; macroparticles kept	0.721–0.827; 0.783; 5.8–9.2 (6.4), 1.8–3.8 (2.4); all	–
Unperturbed: spacing of successive minima of α_c from $200M$, before and after $600M$	44–58; 50–76	maxima 50–55 to 360, 60–70 after (perturbed run; read from figure)
Unperturbed: largest half-swing of α_c between successive extrema (t/M)	0.036 (415)	0.041 (250–275; perturbed run; read from figure)
Unperturbed: α_c range; mean of the last two pairs of successive extrema (t/M); macroparticles kept	0.696–0.772; 0.720 (745), 0.714 (770); all	0.696–0.777; 0.735 (800; perturbed run; read from figure); all kept

radius map take the form

$$
\begin{aligned}
\zeta &= \frac{1-y_c}{3}, \qquad g_y = \frac{y^{-1}-1}{y_c^{-1}-1},\\
\frac{\mathrm{d}y}{\mathrm{d}\tilde r} &= \frac{2\zeta y}{\tilde r^2}\,\frac{\tilde m + \zeta\tilde r^3 (y/y_c)^{-1/2} g_y^{3/2}}{1-2\zeta\tilde m/\tilde r},\\
\frac{\mathrm{d}\tilde m}{\mathrm{d}\tilde r} &= \tilde r^2\left(\frac{y}{y_c}\right)^{-3/2} g_y^{1/2},\\
\frac{\mathrm{d}\tilde m_0}{\mathrm{d}\tilde r} &= \frac{\tilde r^2 y_c^{1/2}(y/y_c)^{-1} g_y^{1/2}}{\sqrt{1-2\zeta\tilde m/\tilde r}},\\
\frac{\mathrm{d}\ln\bar r}{\mathrm{d}\ln\tilde r} &= \frac{1}{\sqrt{1-2\zeta\tilde m/\tilde r}}.
\end{aligned}
\tag{B7}
$$

Near the cluster surface where density vanishes, the variable $\sqrt{1-y}$ keeps every derivative analytic up to $y(\tilde r_{\rm s}) = 1$. Outward integration from $\tilde r = 0$ ($\tilde m = \tilde m_0 = 0$) yields surface values $(\tilde r_{\rm s}, \tilde m_{\rm s}, \tilde m_{0,{\rm s}})$ and interior profiles matched at $\bar r_{\rm s}$ from Eq. (B4),

$$
\begin{aligned}
R &= \frac{M\tilde r_{\rm s}}{\zeta\tilde m_{\rm s}}, \qquad M_0 = \frac{M\tilde m_{0,{\rm s}}}{\tilde m_{\rm s}},\\
E_0 &= \sqrt{1-\frac{2M}{R}}, \qquad \alpha_{\rm eq}(\bar r) = E_0\sqrt{y},\\
\psi_{\rm eq}(\bar r) &= \sqrt{\frac{R\tilde r}{\tilde r_{\rm s}\bar r}},
\end{aligned}
\tag{B8}
$$

with enclosed masses $m_{\rm g} = M\tilde m\tilde m_{\rm s}^{-1}$, $m_0 = M_0\tilde m_0\tilde m_{0,{\rm s}}^{-1}$, and $4\pi\bar r^2\rho_0\psi_{\rm eq}^6\,\mathrm{d}\bar r = \mathrm{d}m_0$. For N equal-mass macroparticles ($p = 1,\dots,N$) and five coordinates $d \in \{1,\dots,5\}$, randomized stratification draws

$$
U_{pd} = \frac{\pi_d(p) + \xi_{pd}}{N},
\tag{B9}
$$

where independent random permutations π_d map $\{1,\dots,N\}$ onto $\{0,\dots,N-1\}$ and ξ_{pd} is uniform on $[0,1)$. The cumulative rest-mass fraction $F(\bar r) = m_0(\bar r)M_0^{-1}$ sets each radial coordinate through $\bar r_p = F^{-1}(U_{p1})$. Position angles $(\cos\theta_p, \phi_p) = (2U_{p2} - 1, 2\pi U_{p3})$ define the radial unit vector $\hat{\boldsymbol r}_p$ and $\boldsymbol x_p = \bar r_p\hat{\boldsymbol r}_p$, while momentum angles $(\cos\theta_{u,p}, \phi_{u,p}) = (2U_{p4} - 1, 2\pi U_{p5})$ define an isotropic unit vector $\hat{\boldsymbol u}_p$. At radius

$\bar{r}_p$, the monoenergetic condition $E_0 = \alpha_{\rm eq}(\bar{r}_p)W_p$ and Eq. (5) fix the local orthonormal momentum magnitude $u_{\rm eq}(\bar{r}_p)$ and covariant momentum,

$$u_{\rm eq}(\bar{r}) = \sqrt{\frac{E_0^2}{\alpha_{\rm eq}(\bar{r})^2} - 1},$$
$$u_{i,p} = \psi_{\rm eq}(\bar{r}_p)^2 u_{\rm eq}(\bar{r}_p)\hat{u}_{i,p}. \tag{B10}$$

Plain Monte Carlo sampling, used as the alternative in the particle-number study, replaces U_{pd} in Eq. (B9) with independent uniform draws on $[0, 1)$. Because the mass-weighted mean momentum of a finite sample does not vanish and makes the cluster drift, that mean is subtracted from every particle,

$$u_{i,p} \leftarrow u_{i,p} - \frac{\sum_{p'=1}^{N} m_{p'} u_{i,p'}}{\sum_{p'=1}^{N} m_{p'}}. \tag{B11}$$

Along the equilibrium sequence, an inward velocity kick of fractional amplitude δ_v scales the radial momentum component as in Ref. [21], shifting $\hat{u}_{i,p}$ in Eq. (B10) to $\hat{u}_{i,p} - \delta_v \hat{r}_{i,p}$ before Eq. (B11). For the homologous density perturbation δ_ρ of the near-critical model, equilibrium radii $\bar{r}_{{\rm eq},p}$ scale by a common factor $\Lambda_{\rm h}$ to $\bar{r}_p = \Lambda_{\rm h}\bar{r}_{{\rm eq},p}$ with $\boldsymbol{x}_p = \bar{r}_p\hat{\boldsymbol{r}}_p$. Unlike a perturbation applied on the unperturbed metric $\psi_{\rm eq}$, here $\psi(\bar{r})$ is recomputed from the perturbed matter by a radial fixed-point solve of the Hamiltonian constraint,

$$\Lambda_{\rm h} = (1+\delta_\rho)^{-1/3},$$
$$\frac{1}{\bar{r}^2}\frac{\mathrm{d}}{\mathrm{d}\bar{r}}\left(\bar{r}^2\frac{\mathrm{d}\psi}{\mathrm{d}\bar{r}}\right) = -\frac{2\pi}{\Lambda_{\rm h}^3\psi(\bar{r})}\,\rho_0\!\left(\frac{\bar{r}}{\Lambda_{\rm h}}\right)\psi_{\rm eq}\!\left(\frac{\bar{r}}{\Lambda_{\rm h}}\right)^6\langle W(\bar{r})\rangle. \tag{B12}$$

Outside $\Lambda_{\rm h}\bar{r}_{\rm s}$, $\psi(\bar{r})$ joins $1 + M'(2\bar{r})^{-1}$ for perturbed ADM mass M', while the interior lapse rescales $\alpha_{\rm eq}(\bar{r}\Lambda_{\rm h}^{-1})$ continuously to the exterior isotropic Schwarzschild lapse. Particle momenta at $\bar{r}_p$ are set from E_0 via $u_{\rm eq}(\bar{r}_{{\rm eq},p})$ lowered with $\psi_{\rm eq}(\bar{r}_p)^2$ before Eq. (B11), while $\langle W(\bar{r})\rangle$ averages Eq. (5) over directions in the new metric $\psi(\bar{r})^4\delta_{ij}$. An inward velocity kick with vanishing extrinsic curvature $K_{ij} = 0$ leaves a first-order violation of the momentum constraint $\mathcal{M}_i = 0$, which is not corrected.

The relativistic cluster with two collapsing overdensities superposes a monoenergetic envelope and two half-turn-symmetric cold Gaussian components on an equatorial circular orbit of radius $d_{\rm orb}$. At orbital radius $d_{\rm orb}$ enclosing envelope mass $m_{\rm g}(d_{\rm orb})$, leading-order circular motion gives

$$v_{\rm orb}^2 = \frac{m_{\rm g}(d_{\rm orb})}{d_{\rm orb}} + \frac{m_c}{4d_{\rm orb}}, \tag{B13}$$

which matches the speed quoted in Ref. [21] only for a host cluster less than half as large as the stated radius. The cluster size in the published figure agrees with that smaller value, and the rebuild adopts this self-consistent host radius in the parameter table.

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